A Stabilized Vector Nodal Meshless Method for $H(\mathbf{curl})$ 2D Electromagnetic Problems on Irregular Point Clouds

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Abstract

The Vector Nodal Meshless Method (VNMM) provides a node-based framework for $H(\mathbf{curl})$ vector-field approximation on point clouds. While early formulations relied on structured background subdivisions, extending the method to unconstrained point clouds reveals that the conventional incomplete solenoidal space ($\mathcal{L}^1$) triggers a persistent $\mathcal{O}(1)$ error plateau in curl approximation. This work resolves this limitation through a two-level framework addressing local polynomial consistency and global variational stability. Locally, $\mathcal{L}^1$ is replaced by the complete linear space $\mathcal{P}_1 \times \mathcal{P}_1$, capturing the full field Jacobian, eliminating derivative aliasing, and restoring $\mathcal{O}(h)$ curl convergence. Degenerate stencils are rejected through scale-aware determinant qualification on a dimensionless placement matrix $\widehat{\mathbf{A}}$ coupled with k-d tree searches. Globally, we prove that while VNMM exhibits an exact constructive equivalence to lowest-order Whitney edge elements on conforming simplices, dynamic support selection on unconstrained clouds produces a globally non-conforming approximation lacking discrete de Rham sequence exactness. This causes discrete gradient nullspace leakage in curl-curl eigenvalue formulations. We resolve this by incorporating a div-curl regularized Ritz-Galerkin

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formulation that shifts spurious modes away from the operational spectrum, leaving physical cavity resonances unperturbed. The findings establish that integrating complete vector polynomials, scale-invariant support selection, and variational div-curl regularization provides a robust, stabilized meshless framework for computational electromagnetics across regular and irregular point clouds.

Keywords: Meshless methods, Vector Nodal Meshless Method (VNMM), $H(\mathbf{curl})$ approximation, Spurious modes, Resonant cavities, Electromagnetics

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1. Introduction

1.1. Context and Motivation

The numerical simulation of Maxwell's equations is central to modern computational electromagnetics (CEM) [1, 2, 3]. In CEM, discretizing Maxwell's equations via conventional node-based scalar Finite Element Methods (FEM) is well known to trigger severe pathologies [1, 4]: artificial locking of normal discontinuities across material boundaries, inaccurate modeling of edge singularities, and spectral pollution by non-physical spurious modes resulting from the violation of the discrete de Rham complex ($\mathrm{Im}(\nabla) \not\subset \ker(\nabla\times)$) [5, 6, 7]. Although Nédélec's edge finite elements (EFEM/Whitney forms) [8, 3] resolve these issues by natively conforming in $H(\mathbf{curl};\Omega)$, they require high-quality conforming meshes. For complex CAD geometries, moving parts in electrical machinery, and adaptive refinement, mesh generation and distortion repair represent severe computational bottlenecks [9, 10, 11].

To circumvent mesh constraints, meshless methods construct field approximations directly on unstructured point clouds [10, 12]. Early adaptations based on scalar Moving Least Squares (EFG) or local Petrov-Galerkin schemes (MLPG) [13, 14, 15, 16, 17] approximated Cartesian components (E_x, E_y) independently, re-introducing spurious modes and tangential locking unless augmented by costly divergence-free projections or Lagrange multipliers [18, 19]. To provide native vector regularity, Lima and Mesquita introduced the Edge Meshless Method (EMM) [20], subsequently equipped by Silva et al. [21] with Nédélec-type $\mathcal{L}^1$ vector shape functions. Taking the asymptotic limit as physical edge lengths contract to point-collocated nodes

($L_k \to 0$), Ortiz et al. [22, 23, 24] established the Vector Nodal Meshless Method (VNMM), later extended to 3D and periodic problems [25, 26, 27].

In the VNMM, the 1D continuous edge is replaced by a vector node $(P_i, \vec{t}_i)$ endowed with a unit director $\vec{t}_i$ [23, 24]. Assigning a single directional projection degree of freedom ($e_i = \vec{E}(P_i) \cdot \vec{t}_i$) per node and aligning boundary directors tangent to interfaces enables natural nodal collocation of tangential continuity and PEC conditions ($e_i = 0$), preserving single-DoF scalar efficiency while retaining local vector regularity [23, 26]. Table 1 compares conventional CEM discretizations with the stabilized VNMM developed herein.

Table 1: Conceptual comparison of spatial discretization paradigms in computational electromagnetics.

Method	**Mesh Req.**	**DoF Type**	H**(curl) Trace**	**Spurious Control**	**Point-Cloud Curl Rate**
Edge FEM (EFEM) [8, 3, 7]	Conformal mesh	Edge circulation $\int_{E_k} \vec{E} \cdot d\boldsymbol{\ell}$	Natively conforming ($\mathbf{n} \times [\vec{E}] = \mathbf{0}$)	Exact discrete de Rham complex	$\mathcal{O}(h^1)$ (simplex-based)
Scalar EFG / MLS [12, 14]	None (Cloud)	Component splitting (E_x, E_y)	Artificial (C^0 normal locking / penalty)	Divergence penalty	$\mathcal{O}(h^p)$ (high-order MLS)
Vector MLPG [15, 17]	None (Local subdomains)	Component splitting (E_x, E_y)	Lagrange multipliers / penalty	Divergence-free projection	$\mathcal{O}(h^p)$ (local MLS integration)
RBF-FD / RPIM [28, 11]	None (Cloud)	Component splitting (E_x, E_y)	Collocation / ghost penalty	Divergence cleaning / penalty	$\mathcal{O}(h^p)$ (radial basis differentiation)
Original VNMM ($\mathcal{L}^1$) [23, 24]	Planar / Delaunay subdivision	Projection $e_k = \vec{E}(P_k) \cdot \vec{t}_k$	Natural tangential alignment	Conditional (structured supports)	$\mathcal{O}(h^1)$ on meshes; $\mathcal{O}(1)$ plateau on clouds
Proposed VNMM ($\mathcal{P}^1$)	None (Arbitrary point clouds)	Projection $e_k = \vec{E}(P_k) \cdot \vec{t}_k$	Natural tangential alignment	Divergence penalty	Consistent $\mathcal{O}(h^1)$ via full Jacobian space

1.2. The Curl Stagnation Bottleneck in Arbitrary Supports

Despite the mathematical elegance of the original VNMM [23, 24], early formulations achieved convergence by relying on structured support selections (e.g., macro-element subdivisions or Delaunay triangulations), where closed-contour circulation cancellation prevented curl error stagnation. However, reliance on underlying meshes undermines the core meshless premise of unconstrained point-cloud discretization [10, 26].

Under automated stencil selection (k-d tree queries) on unconstrained point clouds, the incomplete solenoidal basis $\mathcal{L}^1$ breaks down: **the numerical curl $(\nabla \times \vec{E}^h)_z$ fails to converge, exhibiting an $\mathcal{O}(1)$ error plateau under spatial refinement** (Section 2.2.5). Unlike conforming edge elements where Stokes' theorem guarantees boundary circulation cancellation [8, 3], point collocation lacks closed-contour integration. In $\mathcal{L}^1$, omitting dilatational gradients ($\partial E_x/\partial x$, $\partial E_y/\partial y$) prevents shape functions from capturing the complete first-order variation; on arbitrary clouds, unrepresented modes contaminate the rotational degree of freedom through $\mathbf{A}^{-1}$, producing persistent modal aliasing.

Furthermore, dynamic support stencils render the global meshless space non-conforming ($V_h \not\subset H(\mathbf{curl}; \Omega)$), breaking discrete de Rham exactness and leaking gradient nullspaces into the non-zero spectrum as spurious modes. Stabilizing the VNMM on unconstrained clouds therefore requires a two-level approach: completing the local polynomial basis ($\mathcal{L}^1 \to \mathcal{P}^1$) to resolve the local Jacobian, and employing a div-curl regularized Ritz-Galerkin weak formulation to restore global spectral purity (Section 4).

1.3. Scope and Main Contributions of the Paper

This paper presents a stabilized Vector Nodal Meshless Method (VNMM) for robust vector field approximations on unconstrained point clouds without mesh connectivity. The core methodological contributions are fourfold:

1. **Elimination of Modal Aliasing and Linear Patch-Test Consistency via $\mathcal{P}^1$:** Adopting the complete linear space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$ resolves all four components of the field Jacobian $\mathbf{J}^h = \nabla \vec{E}^h$ (Section 2.3), eliminating the derivative aliasing responsible for curl stagnation in $\mathcal{L}^1$. Furthermore, it satisfies the Galerkin patch test to machine precision ($\leq 7.4 \times 10^{-14}$) across all affine vector modes—confirming linear completeness of the discrete operator—and delivers systematic asymptotic convergence ($\mathcal{O}(h^{2.08})$ in L^2 and $\mathcal{O}(h^{0.75})$ in $H(\mathbf{curl})$) for driven source

problems via the Method of Manufactured Solutions (MMS), whereas $\mathcal{L}^1$ exhibits errors up to 47% and complete convergence stagnation (Section 5.4).

2. **Scale-Invariant Support Selection via Dimensionless Placement Matrix $\widehat{\mathbf{A}}$:** To enable stable mesh-free operation on unstructured clouds, we characterize matrix singularity in $\mathbb{R}^{6\times 6}$ via the Null-Space Orthogonality Principle (Section 3.2) and eliminate the quartic scaling $\det(\mathbf{A}) \sim \mathcal{O}(h^4)$ by introducing a dimensionless placement matrix $\widehat{\mathbf{A}}$. Normalizing local stencils by the spacing h_{local} decouples qualification ($|\det(\widehat{\mathbf{A}})| \geq \widehat{\text{Tol}} > 0$) from physical dimensions, preserving compact locality without support dilation as $h \to 0$ (Section 3).
3. **Rigorous Local Constructive Equivalence Proof:** We establish the Local Dual Equivalence Theorem (Section 2.2.3), proving that under conforming midpoint/tangent placement, local VNMM $\mathcal{L}^1$ shape functions and lowest-order Whitney 1-forms are constructively identical. This result clarifies the fundamental distinction between mesh-conforming topologies, where shared cell facets guarantee global $H(\mathbf{curl})$ conformity, and unconstrained point clouds, where dynamically shifting stencils inevitably induce global non-conformity.
4. **Variational Div-Curl Regularization for Non-Conforming Meshless Discretizations:** To address gradient leakage stemming from global non-conformity ($V_h \not\subset H(\mathbf{curl};\Omega)$), we formulate a div-curl regularized Ritz-Galerkin weak formulation ($\mathbf{K} = \mathbf{K}_{\text{curl}} + s_{\text{div}}\mathbf{K}_{\text{div}}$, Section 4), reliably shifting non-physical gradient modes outside the operational bandwidth without auxiliary Lagrange multipliers or saddle-point indefiniteness.

The remainder of this paper is organized as follows: Section 2 formulates the VNMM shape functions for the $\mathcal{L}^1$ and $\mathcal{P}^1$ bases and proves the Local Dual Equivalence Theorem; Section 3 analyzes placement singularities and develops scale-invariant support selection; Section 4 establishes the div-curl regularized weak form; Section 5 presents numerical verification (patch test, MMS, cavity spectra, and irregular clouds); and Section 6 concludes the paper.

2. Mathematical Formulations of the 2D VNMM

This section establishes the functional framework of the two-dimensional Vector Nodal Meshless Method (VNMM 2D). We define the single-DoF nodal projection scheme and pointwise boundary conditions, analyze the topological equivalence between the classical $\mathcal{L}^1$ basis and Whitney 1-forms on conforming simplices, and formulate the complete linear vector space $\mathcal{P}^1$ to resolve the local field Jacobian and eliminate derivative aliasing.

2.1. Nodal Projection and Tangential Continuity

The VNMM [23, 24] constructs local $H(\mathbf{curl})$-regular vector approximations by distributing a set of M nodes throughout the computational domain Ω and its boundaries $\partial\Omega$. Unlike traditional scalar meshless formulations that assign separate degrees of freedom to individual spatial components (e.g., E_x and E_y), the VNMM associates a single physical unit vector director $\vec{t}_i = [t_{ix}, t_{iy}]^T$ with each node i.

The approximate vector field $\vec{E}^h(\mathbf{x})$ at any spatial location $\mathbf{x} = (x, y) \in \Omega$ is expressed as a linear combination of vector nodal shape functions $\vec{N}_i(\mathbf{x})$:

$$\vec{E}^h(x, y) = \sum_{i=1}^{M_{\text{supp}}} \vec{N}_i(x, y) e_i \tag{1}$$

where M_{supp} is the number of active support nodes in the local influence domain, and e_i represents the discrete scalar degree of freedom (DoF) associated with the i-th node. Physically, e_i corresponds to the pointwise directional projection of the electric field vector along the unit vector director $\vec{t}_i$ at the node coordinate $\mathbf{x}_i = (x_i, y_i)$:

$$e_i = \vec{E}(\mathbf{x}_i) \cdot \vec{t}_i \quad [\text{Volts/meter}] \tag{2}$$

To ensure physical consistency, local curl regularity, and direct enforcement of boundary conditions, the vector shape functions $\vec{N}_i(\mathbf{x})$ are required to satisfy the vector Kronecker delta projection property [20, 23]:

$$\vec{N}_i(\mathbf{x}_k) \cdot \vec{t}_k = \delta_{ik} = \begin{cases} 1, & \text{if } k = i \\ 0, & \text{if } k \neq i \end{cases} \tag{3}$$

for all nodes k belonging to the local support domain. Property (3) dictates that the shape function $\vec{N}_i$ exhibits a unit projection along its own nodal director $\vec{t}_i$ at position $\mathbf{x}_i$, while maintaining zero projection along the directors of all other surrounding support nodes.

2.1.1. *Enforcement of PEC Boundary Conditions and Remarks on Material Interfaces*

The vector Kronecker delta property (3) enables a direct, unconstrained nodal collocation of essential Dirichlet boundary conditions and tangential field continuity without requiring background auxiliary meshes, penalty parameters, or Lagrange multipliers [25, 23].

For nodes lying on Perfect Electric Conductor (PEC) boundaries Γ_{PEC}, where the physical boundary condition requires $\hat{n} \times \vec{E} = \mathbf{0}$ ($\hat{n}$ being the outward unit normal to $\partial\Omega$), the associated vector director $\vec{t}_k$ is aligned strictly tangent to the boundary contour:

$$\vec{t}_k \parallel \partial\Omega \implies \hat{n}(\mathbf{x}_k) \cdot \vec{t}_k = 0. \tag{4}$$

Consequently, setting the scalar nodal degree of freedom directly to zero:

$$e_k = 0 \quad \forall \mathbf{x}_k \in \Gamma_{\text{PEC}} \tag{5}$$

enforces the PEC requirement pointwise at every boundary node:

$$\vec{E}^h(\mathbf{x}_k) \cdot \vec{t}_k = 0 \quad \forall \mathbf{x}_k \in \Gamma_{\text{PEC}}, \tag{6}$$

by virtue of the vector Kronecker delta property (3).

Nodal Collocation versus Continuous Boundary Trace. From a functional analysis viewpoint, the pointwise nodal collocation condition (6) must be distinguished from the continuous functional trace condition:

$$\hat{n} \times \vec{E}^h = \mathbf{0} \quad \text{everywhere on } \Gamma_{\text{PEC}}. \tag{7}$$

On conforming simplicial triangulations (Section 2.2.3), internal DoFs have identically vanishing tangential traces along boundary edges, so the nodal constraint (6) guarantees that Eq. (7) holds everywhere along Γ_{PEC}. On unconstrained point clouds, however, the global approximation is non-conforming ($V_h \not\subset H(\mathbf{curl}; \Omega)$). For points $\mathbf{x} \in \Gamma_{\text{PEC}}$ between nodes, dynamic stencils $\mathcal{S}(\mathbf{x})$ include interior nodes with $e_i \neq 0$ whose shape functions do not vanish tangentially along $\partial\Omega$. Setting $e_k = 0$ enforces the PEC condition strictly at boundary nodes $\mathbf{x}_k$, while continuous trace satisfaction is variational. Similarly, on curved boundaries with tangent directors $\vec{t}_k = \hat{\tau}(\mathbf{x}_k)$, the condition is collocated directly without polygonization, while the trace between nodes converges under spatial refinement.

Material Interfaces and Homogeneous Scope. Assigning unit directors tangent to material interfaces Γ_{int} ($\vec{t}_i = \hat{\tau}(\mathbf{x}_i)$) enables direct collocation of tangential continuity ($\vec{E}_1^h \cdot \vec{t}_i = \vec{E}_2^h \cdot \vec{t}_i = e_i$) while permitting the normal field to undergo a physical step discontinuity ($E_{n1} \neq E_{n2}$), avoiding the normal locking of scalar C^0 meshless schemes. Normal flux continuity ($D_{n1} = D_{n2} \iff \epsilon_{r1}E_{n1} = \epsilon_{r2}E_{n2}$) is enforced variationally by the weak form and divergence penalty. Rigorous handling of heterogeneous media requires partitioned support stencils (e.g., multi-tree k-d trees [29]) to prevent unphysical gradient smoothing across subdomains; because this work focuses on homogeneous benchmarks ($\epsilon_r = 1, \mu_r = 1$), systematic treatment of heterogeneous interfaces is deferred to future work (Section 6.2).

2.2. The Incomplete $\mathcal{L}^1$ Basis (3 Support Nodes)

The construction of vector shape functions in the 2D VNMM directly inherits the mathematical foundations of $H(\mathbf{curl})$-conforming edge-based meshless approximations [20, 21]. The incomplete 3-term solenoidal polynomial space

$$\mathcal{L}^1 = \text{span}\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}y\\-x\end{bmatrix}\} \tag{8}$$

corresponds to the local two-dimensional reduction of Nédélec's first-type space [8], whose vector projection mechanisms were established for edge meshless line segments by Silva et al. [21] and later adapted to point-collocated nodes [23, 24].

Consequently, the local vector shape function $\vec{N}_i$ associated with the i-th support node in a 3-node local domain is defined as a linear combination of these basis vectors:

$$\vec{N}_i(x, y) = \beta_{1i}\begin{bmatrix}1\\0\end{bmatrix} + \beta_{2i}\begin{bmatrix}0\\1\end{bmatrix} + \beta_{3i}\begin{bmatrix}y\\-x\end{bmatrix} \tag{9}$$

where $\boldsymbol{\beta}_i = [\beta_{1i}, \beta_{2i}, \beta_{3i}]^T \in \mathbb{R}^3$ represents the unknown local interpolation coefficient vector.

To determine these coefficients, the vector shape functions are constrained to satisfy the vector Kronecker delta projection property (3) for $i, k = 1, 2, 3$,

Applying constraint (3) to all three support nodes leads to the local 3×3 linear placement system:

$$\mathbf{A}\boldsymbol{\beta}_i = \mathbf{L}_i \tag{10}$$

where the local moment placement matrix $\mathbf{A} \in \mathbb{R}^{3\times3}$ is explicitly given by:

$$\mathbf{A} = \begin{bmatrix} t_{1x} & t_{1y} & y_1 t_{1x} - x_1 t_{1y} \\ t_{2x} & t_{2y} & y_2 t_{2x} - x_2 t_{2y} \\ t_{3x} & t_{3y} & y_3 t_{3x} - x_3 t_{3y} \end{bmatrix} \tag{11}$$

and the canonical load vectors are $\mathbf{L}_1 = [1, 0, 0]^T$, $\mathbf{L}_2 = [0, 1, 0]^T$, and $\mathbf{L}_3 = [0, 0, 1]^T$.

Once the coefficients $\boldsymbol{\beta}_i = \mathbf{A}^{-1}\mathbf{L}_i$ are evaluated, the approximated electric field vector $\vec{E}^h(x, y)$ is constructed via nodal projection superposition:

$$\vec{E}^h(x, y) = \sum_{i=1}^{3} \vec{N}_i(x, y) e_i \tag{12}$$

where $e_i = \vec{E}(\mathbf{x}_i) \cdot \vec{t}_i$ represent the discrete scalar projection DoFs.

2.2.1. Mathematical Properties

The shape functions $\vec{N}_i$ of the $\mathcal{L}^1$ basis possess two distinct mathematical characteristics:

1. **Identical Solenoidal Property ($\nabla \cdot \vec{N}_i \equiv 0$):** Taking the divergence of Eq. (9) yields:

$$\nabla \cdot \vec{N}_i = \frac{\partial}{\partial x}(\beta_{1i} + \beta_{3i} y) + \frac{\partial}{\partial y}(\beta_{2i} - \beta_{3i} x) \equiv 0 \tag{13}$$

2. **Univariate Curl Representation:** Evaluating the scalar z-directed curl operator on $\vec{N}_i$ yields:

$$(\nabla \times \vec{N}_i)_z = \frac{\partial N_{iy}}{\partial x} - \frac{\partial N_{ix}}{\partial y} = \frac{\partial(-\beta_{3i} x)}{\partial x} - \frac{\partial(\beta_{3i} y)}{\partial y} = -2\beta_{3i} \tag{14}$$

Therefore, the approximated z-directed scalar curl field $(\nabla \times \vec{E}^h)_z$ is spatially constant across the 3-node support domain and depends exclusively on the third coefficient β_{3i}:

$$(\nabla \times \vec{E}^h)_z = -2 \sum_{i=1}^{3} \beta_{3i} e_i \tag{15}$$

2.2.2. Local Constructive Equivalence under Conforming Midpoint/Tangent Placement

The VNMM 2D was originally conceptualized as the zero-edge-length asymptotic limit of the Edge Meshless Method (EMM) [20, 23]. By examining this heritage, a fundamental local constructive equivalence under conforming midpoint/tangent placement between the VNMM point-collocation formulation and classic lowest-order Whitney edge finite elements (Nédélec 1-forms on triangular simplices [8, 3, 7]) is formally established, as detailed in Fig. 1.

In conventional FEM (Fig. 1a), the degrees of freedom associated with the triangular cell Ω_e are line integrals of circulation $\alpha_k = \oint_{e_k} \vec{E} \cdot d\boldsymbol{\ell}$ along continuous 1D edges of finite physical length L_k. As demonstrated in the asymptotic collapse limit (Fig. 1c), when the physical edge length shrinks to zero ($L_k \to 0$), the 1D continuous edge degenerates into a point-collocated vector node $(P_k, \vec{t}_k)$, where the circulation DoF α_k naturally scales into a pointwise directional field projection $e_k = \vec{E}(P_k) \cdot \vec{t}_k$ [Volts/meter].

2.2.3. The Local Dual Equivalence Theorem

Let $T \subset \mathbb{R}^2$ be an arbitrary non-degenerate triangular simplex with vertices V_1, V_2, V_3 ordered counter-clockwise and non-zero area $|T| > 0$. Let $e_k = (V_k, V_{k+1})$ be the directed boundary edges of length $L_k = \|V_{k+1} - V_k\| > 0$, with unit tangent vectors $\vec{t}_k = (V_{k+1} - V_k)/L_k$ and midpoints $M_k = \frac{1}{2}(V_k + V_{k+1})$ for $k \in \{1, 2, 3\}$ (with index wrap-around $V_4 \equiv V_1$).

Theorem 1 (Local Constructive Equivalence under Conforming Midpoint/Tangent Placement)**.** *Consider the support triad of vector nodes* $\{P_k, \vec{t}_k\}_{k=1}^3$ *defined at the edge midpoints* $P_k = M_k$ *with directors* $\vec{t}_k$*. The VNMM 2D* $\mathcal{L}^1$ *shape functions* $\vec{N}_k(\mathbf{x})$ *and the Whitney 1-forms* $\vec{w}_k(\mathbf{x}) = \lambda_i(\mathbf{x})\nabla\lambda_j - \lambda_j(\mathbf{x})\nabla\lambda_i$ *(with* $(i, j) = (k, k+1)$*) satisfy:*

$$\vec{N}_k(\mathbf{x}) \equiv L_k \, \vec{w}_k(\mathbf{x}) \quad \Longleftrightarrow \quad \vec{w}_k(\mathbf{x}) \equiv \frac{1}{L_k} \vec{N}_k(\mathbf{x}), \quad \forall \mathbf{x} \in \mathbb{R}^2, \ \forall k \in \{1, 2, 3\} \tag{16}$$

Outline of Proof. The constructive equivalence follows from two geometric and algebraic properties: (i) both the Whitney 1-form space $\mathcal{W}(T)$ and the VNMM $\mathcal{L}^1$ basis span identical 3-dimensional vector polynomial spaces; and (ii) the tangential projection $\vec{u}(\mathbf{x}) \cdot \vec{t}_k$ of any affine field $\vec{u} \in \mathcal{L}^1$ is strictly invariant along each edge e_k. Consequently, boundary circulation integrals reduce

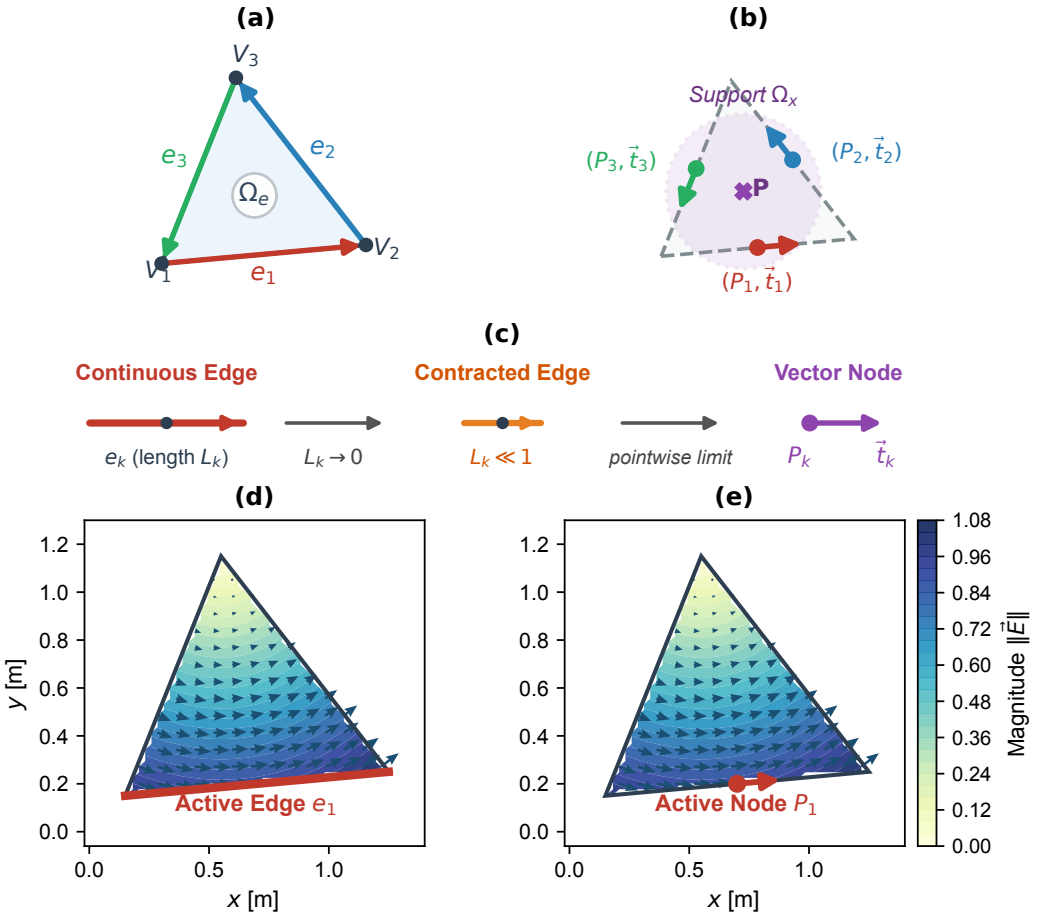


Figure 1: Local equivalence between Whitney edge elements and the VNMM $\mathcal{L}^1$ basis. (a) Triangular cell Ω_e with vertices V_i and oriented edges e_k; (b) VNMM support stencil with vector nodes $(P_k, \vec{t}_k)$ placed at edge midpoints; (c) Asymptotic collapse ($L_k \to 0$) mapping edge circulations to directional projections e_k; (d) Whitney 1-form $\vec{w}_1$ for edge e_1; (e) Normalized VNMM basis $\vec{N}_1/L_1$ for node $(P_1, \vec{t}_1)$, displaying identical vector fields over Ω_e.

to midpoint evaluations scaled by the edge length, $\oint_{e_k} \vec{u} \cdot d\boldsymbol{\ell} = L_k(\vec{u}(M_k) \cdot \vec{t}_k)$, establishing an exact unisolvent isomorphism between Whitney circulations and VNMM point-collocated directional projections ($\vec{N}_k \equiv L_k \vec{w}_k$). The complete five-step constructive proof is provided in Appendix A. □

As graphically illustrated in Fig. 1d and 1e (and verified numerically in Section 5.1), the normalized VNMM $\mathcal{L}^1$ shape functions $\frac{1}{L_k}\vec{N}_k$ and canonical

Whitney 1-forms $\vec{w}_k$ match identically across the simplex T down to floating-point machine precision ($\approx 10^{-16}$).

2.2.4. Local $H(\mathbf{curl})$ Regularity versus Global $H(\mathbf{curl})$ Conformity

Beyond establishing the mathematical connection to edge finite elements, this constructive identity clarifies a fundamental structural distinction in vector meshless approximations:

1. **Local $H(\mathbf{curl})$ Regularity on Virtual Simplices:** In conventional scalar meshless methods (e.g., Moving Least Squares or Element-Free Galerkin), evaluating field approximations outside the physical convex hull enclosed by the node coordinates $\mathbf{x}_k$ constitutes an unconstrained extrapolation, leading to rapid truncation error growth [20]. In contrast, because the VNMM vector basis reproduces the complete polynomial space of Whitney edge elements over the virtual macro-triangle Ω_x defined by the intersecting lines of action of vector directors $\vec{t}_k$ (Fig. 1b), the valid domain for local $H(\mathbf{curl})$ field approximation is not geographically restricted to the small physical convex hull of $\mathbf{x}_k$ [8, 3]. Under regular-support assumptions (well-conditioned local placement matrix $\mathbf{A}$ and non-degenerate intersection vertices), any field evaluation at $P \in \Omega_x$ preserves local $H(\mathbf{curl})$ regularity.
2. **Loss of Global $H(\mathbf{curl})$ Conformity on Unconstrained Point Clouds:** When nodes and directors form a conforming simplicial mesh (Section 2.2.3), tangential traces across adjacent element facets match identically, yielding global conformity $V_h \subset H(\mathbf{curl}; \Omega)$. On unconstrained point clouds, however, dynamically shifting support stencils induce inter-stencil tangential trace discontinuities, rendering the global approximation non-conforming ($V_h \not\subset H(\mathbf{curl}; \Omega)$). The spectral consequences of this global non-conformity and its variational stabilization are addressed in Section 4.

2.2.5. Mathematical Origin of Modal Aliasing and Curl Stagnation

While the incomplete solenoidal basis $\mathcal{L}^1$ exhibits exact dual equivalence to Whitney elements on conforming meshes (Section 2.2.2), its behavior diverges sharply on unconstrained point clouds, where curl approximations stagnate at an $\mathcal{O}(1)$ relative error plateau under spatial refinement (as numerically confirmed in Section 5.2.2 and Fig. 5b). To uncover the algebraic origin of this derivative stagnation, we examine the local Taylor series expansion of the exact electric field $\vec{E}$ centered at an evaluation point $P(x_P, y_P)$

[30, 10]:

$$\vec{E}(\mathbf{x}) = \vec{E}(P) + \mathbf{J}(P)\begin{bmatrix}\Delta x\\ \Delta y\end{bmatrix} + \mathcal{O}(h^2) \tag{17}$$

where $\mathbf{J}(P) = \nabla\vec{E}$ represents the complete first-order field Jacobian tensor [10]:

$$\mathbf{J} = \begin{bmatrix}\frac{\partial E_x}{\partial x} & \frac{\partial E_x}{\partial y}\\ \frac{\partial E_y}{\partial x} & \frac{\partial E_y}{\partial y}\end{bmatrix} \tag{18}$$

By expressing individual spatial derivatives in terms of fundamental vector calculus operators—namely, the field divergence $\nabla\cdot\vec{E} = \partial E_x/\partial x + \partial E_y/\partial y$ and the z-directed field curl $(\nabla\times\vec{E})_z = \partial E_y/\partial x - \partial E_x/\partial y$—the field Jacobian tensor is decomposed as:

$$\mathbf{J} = \frac{1}{2}(\nabla\cdot\vec{E})\mathbf{I} + \frac{1}{2}(\nabla\times\vec{E})_z\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix} + \begin{bmatrix}\delta_{\text{norm}} & \delta_{\text{trans}}\\ \delta_{\text{trans}} & -\delta_{\text{norm}}\end{bmatrix} \equiv \mathbf{J}_{\text{div}} + \mathbf{J}_{\text{rot}} + \mathbf{J}_{\text{shear}} \tag{19}$$

where the three matrix terms represent, respectively: (i) the **isotropic dilation (divergence) matrix** $\mathbf{J}_{\text{div}} = \frac{1}{2}(\nabla\cdot\vec{E})\mathbf{I}$, with $\mathbf{I}\in\mathbb{R}^{2\times 2}$ being the identity matrix; (ii) the **skew-symmetric rotational (curl) matrix** $\mathbf{J}_{\text{rot}} = \frac{1}{2}(\nabla\times\vec{E})_z\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$; and (iii) the **symmetric trace-free shear (deviatoric) matrix** $\mathbf{J}_{\text{shear}} = \begin{bmatrix}\delta_{\text{norm}} & \delta_{\text{trans}}\\ \delta_{\text{trans}} & -\delta_{\text{norm}}\end{bmatrix}$ (satisfying $\text{tr}(\mathbf{J}_{\text{shear}}) = 0$). Here, $\delta_{\text{norm}} = \frac{1}{2}(\partial E_x/\partial x - \partial E_y/\partial y)$ denotes the normal (longitudinal) field derivative difference, and $\delta_{\text{trans}} = \frac{1}{2}(\partial E_y/\partial x + \partial E_x/\partial y)$ represents the symmetric transverse (shear) field derivative component.

In the incomplete $\mathcal{L}^1$ basis, local shape functions provide only 3 degrees of freedom. As a result, normal field derivatives are artificially forced to vanish identically ($\partial E_x^h/\partial x \equiv 0$ and $\partial E_y^h/\partial y \equiv 0$), imposing $\nabla\cdot\vec{E}^h \equiv 0$, $\delta_{\text{norm}} \equiv 0$, and $\delta_{\text{trans}} \equiv 0$ simultaneously.

This fundamental limitation is captured by the following order-of-magnitude scaling heuristic:

Heuristic 1 (Jacobian Incompleteness and Derivative Modal Aliasing Scaling)**.** *Let $\vec{E} \in [C^2(\Omega)]^2$ be an arbitrary twice continuously differentiable 2D vector field. The incomplete solenoidal space $\mathcal{L}^1 = \text{span}\{[1,0]^T, [0,1]^T, [y,-x]^T\} \subset \mathcal{P}_1\times\mathcal{P}_1$ cannot reproduce the complete first-order Taylor expansion of $\vec{E}$. Specifically, while the complete space*

$\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$ *possesses* $\dim(\mathcal{P}^1) = 6$ *degrees of freedom to reconstruct the 2 translation components and all 4 independent entries of the Jacobian tensor* $\mathbf{J} = \nabla\vec{E}$, $\mathcal{L}^1$ *possesses only* $\dim(\mathcal{L}^1) = 3$ *degrees of freedom, enforcing the artificial linear constraints:*

$$\frac{\partial E_x^h}{\partial x} = 0, \quad \frac{\partial E_y^h}{\partial y} = 0, \quad \textit{and} \quad \frac{\partial E_x^h}{\partial y} + \frac{\partial E_y^h}{\partial x} = 0. \tag{20}$$

On unconstrained point clouds lacking conforming circulation cancellation, projecting the unrepresented Jacobian components onto the 3-node stencil induces a residual $\|\mathbf{r}\| \sim \mathcal{O}(h)$ *which, under the* $\mathcal{O}(h^{-1})$ *scaling of the rotational row of* $\mathbf{A}^{-1}$*, produces an* $\mathcal{O}(1)$ *derivative aliasing error:*

$$|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z| \sim \| \operatorname{Row}_3(\mathbf{A}^{-1})\| \|\mathbf{r}\| = \mathcal{O}(h^{-1}) \cdot \mathcal{O}(h) = \mathcal{O}(1). \tag{21}$$

The algebraic derivation of this scaling estimate is detailed in Appendix B.

Remark 1 (Scope of Heuristic 1 and the Conforming Case)**.** *The estimate in Heuristic 1 is an order-of-magnitude scaling heuristic, attained when the weights of* $\operatorname{Row}_3(\mathbf{A}^{-1})$ *fail to annihilate the unrepresented symmetric component* $\mathbf{S} = \mathbf{J}_{\text{div}} + \mathbf{J}_{\text{shear}} = \frac{1}{2}(\mathbf{J} + \mathbf{J}^T)$ *of the Jacobian tensor. In the conforming midpoint/tangent configuration of Theorem 1,* $\operatorname{Row}_3(\mathbf{A}^{-1})_k \propto L_k/|T|$ *and the residual satisfies* $\sum_k L_k r_k = \oint_{\partial T}[(\mathbf{J}_{\text{div}} + \mathbf{J}_{\text{shear}})\mathbf{x}] \cdot d\boldsymbol{\ell} = \int_T \nabla \times (\mathbf{S}\mathbf{x})\, d\Omega = 0$*, since* $\sum_k L_k \vec{t}_k = \mathbf{0}$*, the midpoint rule is exact for linear integrands, and both* $\mathbf{J}_{\text{div}}$ *and* $\mathbf{J}_{\text{shear}}$ *are symmetric (ensuring* $\nabla \times (\mathbf{J}_{\text{div}}\mathbf{x}) = \mathbf{0}$ *and* $\nabla \times (\mathbf{J}_{\text{shear}}\mathbf{x}) = \mathbf{0}$*). The residual therefore does not reach the curl degree of freedom, consistently with the* $\mathcal{O}(h)$ *curl error of Whitney elements. The* $\mathcal{O}(1)$ *plateau should thus be read as the generic behaviour on arbitrary node/director placements where this closed-contour cancellation is absent; it represents an order-of-magnitude heuristic confirmed numerically across unconstrained clouds (Section 5.2) rather than a uniform mathematical lower bound for all point configurations.*

Consequently, the $\mathcal{O}(1)$ derivative noise floor on unconstrained point clouds cannot be eliminated by spatial refinement alone ($h \to 0$). Adopting the complete linear polynomial vector space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$, formulated next in Section 2.3, provides a direct remedy by resolving the complete local Jacobian.

2.3. The Complete $\mathcal{P}^1$ Vector Polynomial Basis

To eliminate the derivative modal aliasing and curl stagnation that occur when combining the incomplete $\mathcal{L}^1$ space with automated support selection on unconstrained point clouds, we reformulate local vector interpolation using the complete linear polynomial vector space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$. This transition draws a direct mathematical analogy to Nédélec's second family of mixed finite elements [31], in which vector field approximations are constructed over complete polynomial spaces $(\mathbb{P}_k)^d$ rather than solenoidal-restricted subspaces.

In two dimensions ($d = 2$), a complete first-order vector polynomial basis $\mathcal{P}^1$ provides 6 independent degrees of freedom ($M_{\text{supp}} = 6$ support nodes per evaluation point):

$$\mathcal{P}^1 = \text{span}\left\{ \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}x\\0\end{bmatrix}, \begin{bmatrix}y\\0\end{bmatrix}, \begin{bmatrix}0\\x\end{bmatrix}, \begin{bmatrix}0\\y\end{bmatrix} \right\} \tag{22}$$

Using this complete basis, the local vector shape function $\vec{N}_i(\mathbf{x})$ associated with the i-th support node is expressed as a linear combination of its six constituent terms:

$$\vec{N}_i(x,y) = \beta_{1i}\begin{bmatrix}1\\0\end{bmatrix} + \beta_{2i}\begin{bmatrix}0\\1\end{bmatrix} + \beta_{3i}\begin{bmatrix}x\\0\end{bmatrix} + \beta_{4i}\begin{bmatrix}y\\0\end{bmatrix} + \beta_{5i}\begin{bmatrix}0\\x\end{bmatrix} + \beta_{6i}\begin{bmatrix}0\\y\end{bmatrix} \tag{23}$$

where β_{ji} (for $j = 1, \ldots, 6$) represent the local interpolation coefficients to be determined.

Enforcing the vector Kronecker delta projection property [23]:

$$\vec{N}_i(\mathbf{x}_k) \cdot \vec{t}_k = \delta_{ik}, \quad \text{for } k = 1, \ldots, 6 \tag{24}$$

across all six support nodes yields the local linear placement system

$$\mathbf{A}\boldsymbol{\beta}_i = \mathbf{L}_i. \tag{25}$$

Evaluated in local translated coordinates centered at the evaluation point $P(x_P, y_P)$ (defined as $\Delta x_k = x_k - x_P$ and $\Delta y_k = y_k - y_P$), the local moment placement matrix $\mathbf{A} \in \mathbb{R}^{6\times 6}$ is explicitly given by:

$$\mathbf{A} = \begin{bmatrix} t_{1x} & t_{1y} & \Delta x_1 t_{1x} & \Delta y_1 t_{1x} & \Delta x_1 t_{1y} & \Delta y_1 t_{1y} \\ t_{2x} & t_{2y} & \Delta x_2 t_{2x} & \Delta y_2 t_{2x} & \Delta x_2 t_{2y} & \Delta y_2 t_{2y} \\ t_{3x} & t_{3y} & \Delta x_3 t_{3x} & \Delta y_3 t_{3x} & \Delta x_3 t_{3y} & \Delta y_3 t_{3y} \\ t_{4x} & t_{4y} & \Delta x_4 t_{4x} & \Delta y_4 t_{4x} & \Delta x_4 t_{4y} & \Delta y_4 t_{4y} \\ t_{5x} & t_{5y} & \Delta x_5 t_{5x} & \Delta y_5 t_{5x} & \Delta x_5 t_{5y} & \Delta y_5 t_{5y} \\ t_{6x} & t_{6y} & \Delta x_6 t_{6x} & \Delta y_6 t_{6x} & \Delta x_6 t_{6y} & \Delta y_6 t_{6y} \end{bmatrix} \tag{26}$$

The corresponding coefficient vector $\boldsymbol{\beta}_i$ and canonical right-hand side vector $\mathbf{L}_i$ are structured as:

$$\boldsymbol{\beta}_i = \begin{bmatrix} \beta_{1i} \\ \beta_{2i} \\ \beta_{3i} \\ \beta_{4i} \\ \beta_{5i} \\ \beta_{6i} \end{bmatrix}, \qquad \mathbf{L}_i = \begin{bmatrix} \delta_{1i} \\ \delta_{2i} \\ \delta_{3i} \\ \delta_{4i} \\ \delta_{5i} \\ \delta_{6i} \end{bmatrix} \tag{27}$$

Solving the linear system (25) via $\boldsymbol{\beta}_i = \mathbf{A}^{-1}\mathbf{L}_i$ completely defines the vector shape functions $\vec{N}_i(\mathbf{x})$. Consequently, the approximated electric field $\vec{E}^h(\mathbf{x})$ and its z-directed curl $(\nabla \times \vec{E}^h)_z$ within the support domain are expressed analytically as:

$$\vec{E}^h(x,y) = \sum_{i=1}^{6} \vec{N}_i(x,y) e_i \tag{28}$$

$$(\nabla \times \vec{E}^h)_z(x,y) = \frac{\partial E_y^h}{\partial x} - \frac{\partial E_x^h}{\partial y} = \sum_{i=1}^{6} (\beta_{5i} - \beta_{4i}) e_i \tag{29}$$

where $e_i = \vec{E}(P_i) \cdot \vec{t}_i$ denotes the scalar directional projection degree of freedom associated with the i-th node.

By providing 6 independent degrees of freedom, the complete $\mathcal{P}^1$ space resolves all four spatial derivative components of the local Jacobian tensor $\mathbf{J}^h = \nabla \vec{E}^h$ simultaneously:

$$\mathbf{J}^h = \begin{bmatrix} \frac{\partial E_x^h}{\partial x} & \frac{\partial E_x^h}{\partial y} \\ \frac{\partial E_y^h}{\partial x} & \frac{\partial E_y^h}{\partial y} \end{bmatrix} = \sum_{i=1}^{6} \begin{bmatrix} \beta_{3i} & \beta_{4i} \\ \beta_{5i} & \beta_{6i} \end{bmatrix} e_i \tag{30}$$

Unlike the incomplete $\mathcal{L}^1$ space, which artificially forces longitudinal (normal) field derivatives ($\partial E_x/\partial x$, $\partial E_y/\partial y$) to vanish, the complete $\mathcal{P}^1$ basis resolves both rotational and irrotational (longitudinal flux variation) components simultaneously. This shields the field curl calculation from longitudinal derivative truncation residues, removing the derivative modal aliasing mechanism and providing the polynomial consistency required for first-order curl convergence, while yielding an observed convergence rate of $\mathcal{O}(h^{2.17})$ for electric field interpolation in the investigated benchmark (Section 5.2).

3. Algebraic Conditioning and Support Selection

The construction of vector nodal shape functions $\vec{N}_i$ in the Vector Nodal Meshless Method (VNMM) relies on solving the local linear placement system $\mathbf{A}\boldsymbol{\beta}_i = \mathbf{L}_i$ (Eq. (25)). A fundamental prerequisite for the existence, uniqueness, and numerical stability of these vector shape functions is that the local moment placement matrix $\mathbf{A}$ remains strictly non-singular ($|\det(\mathbf{A})| > 0$).

Unlike scalar meshless formulations where matrix singularity is dictated purely by spatial nodal distributions (e.g., node co-location or spatial co-linearity), the conditioning of $\mathbf{A}$ in the VNMM is intrinsically coupled to both the spatial coordinates $\mathbf{x}_k = (x_k, y_k)$ and the orientation of the unit vector directors $\vec{t}_k = [t_{kx}, t_{ky}]^T$ assigned to each support node. In the following subsections, the geometric singularity modes of the 3-node basis $\mathcal{L}^1$ are reviewed (Section 3.1), while Section 3.2 establishes the Null-Space Orthogonality Principle governing the complete 6-node basis $\mathcal{P}^1$, providing the mathematical foundation for the automated support selection strategy developed in Section 3.3.1.

3.1. Singularities and Geometric Constraints in the Incomplete $\mathcal{L}^1$ Basis (3×3 Matrix)

To establish a systematic classification of algebraic singularities in the original VNMM 2D formulation [23, 24], we revisit the 3×3 local moment placement matrix $\mathbf{A}$ generated by the 3-node solenoidal basis $\mathcal{L}^1$ (Eq. (11)). Algebraic singularity ($\det(\mathbf{A}) = 0$) occurs when the spatial-directional equations contributed by the candidate triad become linearly dependent, falling into four distinct pathological configurations (Figs. 2a–2d):

1. **Global Director Parallelism (Fig. 2a):** If directors are mutually parallel ($\vec{t}_1 \parallel \vec{t}_2 \parallel \vec{t}_3$), $\mathrm{Col}_1(\mathbf{A}) \propto \mathrm{Col}_2(\mathbf{A})$, yielding $\det(\mathbf{A}) \equiv 0$ irrespective of node positions.
2. **Concurrent Lines of Action (Fig. 2b):** If director lines of action intersect at a common point $P_c(x_c, y_c)$, $\mathrm{Col}_3 = y_c\,\mathrm{Col}_1 - x_c\,\mathrm{Col}_2$, forcing $\det(\mathbf{A}) = 0$.
3. **Overlapping Lines of Action (Fig. 2c):** Collinear nodes sharing a common director orientation produce proportional matrix rows ($\mathrm{rank}(\mathbf{A}) \leq 2$).
4. **Nodal Coincidence (Fig. 2d):** Co-located nodes ($\|\mathbf{x}_i - \mathbf{x}_j\| \to 0$) collapse the spatial moment span.

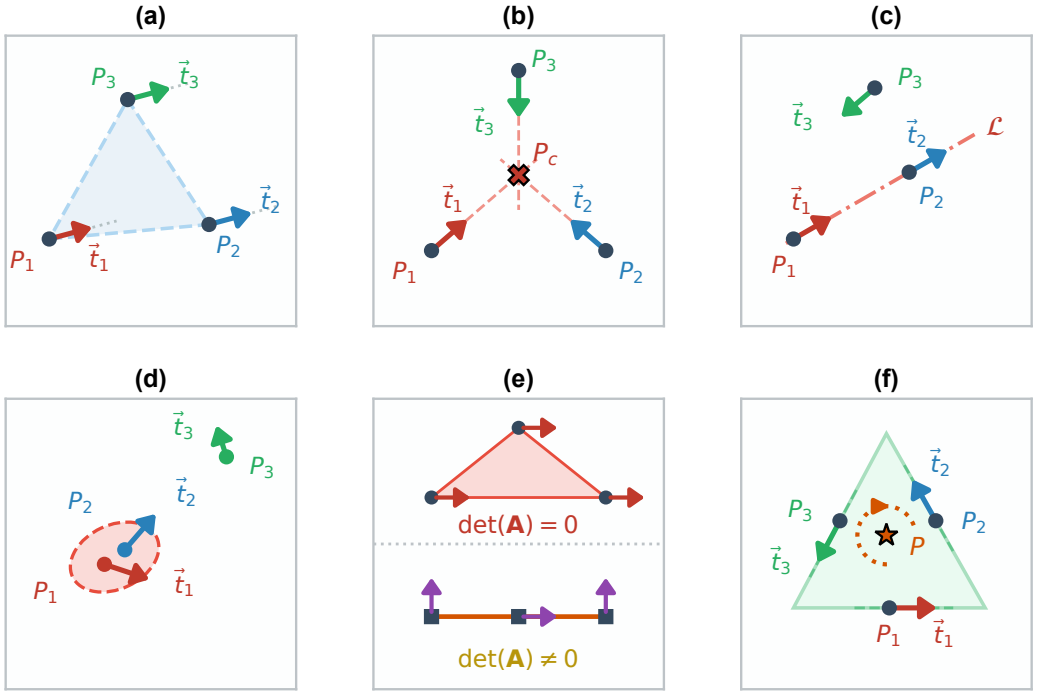


Figure 2: Singularity configurations and support qualification for the VNMM $\mathcal{L}^1$ placement matrix $\mathbf{A}$. (a) Global vector parallelism ($\vec{t}_1 \parallel \vec{t}_2 \parallel \vec{t}_3$); (b) Concurrent lines of action intersecting at focal point P_c; (c) Collinear nodes sharing a common line of action; (d) Nodal coincidence ($\|P_1 - P_2\| \to 0$); (e) Decoupling of nodal geometry and director alignment: spatially non-collinear nodes with parallel directors yield $\det(\mathbf{A}) = 0$ (top), whereas collinear nodes with non-parallel directors fail to span 2D derivatives (bottom); (f) Admissible support: non-collinear distribution ($\text{Area}(T) > 0$) with non-vanishing circulation around evaluation point P.

As illustrated in Fig. 2e, spatial sampling quality is formally decoupled from vector algebraic independence: nodes forming an ideal spatial triangle are singular if directors are parallel ($\det(\mathbf{A}) = 0$, top), whereas collinear nodes can yield $\det(\mathbf{A}) \neq 0$ if directors are non-parallel but fail to span 2D derivatives (bottom). Consequently, an admissible support triad requires dual qualification (Fig. 2f): non-collinear nodal geometry ($|T| > 0$) combined with coherent angular circulation around P.

3.2. Singularity Characterization via the Null-Space Orthogonality Principle

Extending the singularity analysis to the complete vector basis $\mathcal{P}^1$ ($M_{\text{supp}} = 6$), we consider the corresponding 6×6 local placement matrix

$\mathbf{A} \in \mathbb{R}^{6\times 6}$ defined in Eq. (26). In local translated coordinates centered at the evaluation point $P(x_P, y_P)$ ($\Delta x_k = x_k - x_P, \Delta y_k = y_k - y_P$), any coefficient vector $\mathbf{p} = [a_1, a_2, b_1, b_2, b_3, b_4]^T \in \mathbb{R}^6$ uniquely defines an affine vector field $\vec{E}(\mathbf{x}) \in \mathcal{P}^1$:

$$\vec{E}(\mathbf{x}) = \begin{bmatrix} a_1 + b_1\Delta x + b_2\Delta y \\ a_2 + b_3\Delta x + b_4\Delta y \end{bmatrix}. \tag{31}$$

By construction, the k-th row of the homogeneous placement system $\mathbf{Ap} = \mathbf{0}$ computes the pointwise directional projection $\vec{E}(\mathbf{x}_k)\cdot\vec{t}_k = 0$. This establishes the **Null-Space Orthogonality Principle**:

$$\mathbf{p} \in \ker(\mathbf{A}) \iff \vec{E}(\mathbf{x}_k)\cdot\vec{t}_k = 0, \quad \forall k \in \{1,\dots,6\}. \tag{32}$$

Consequently, exact algebraic singularity ($\det(\mathbf{A}) = 0$) occurs if and only if there exists a non-trivial affine vector field $\vec{E} \in \mathcal{P}^1 \setminus \{\mathbf{0}\}$ that is pointwise orthogonal to all six nodal directors at their respective spatial positions.

In the 24-dimensional configuration space of support node coordinates and director orientations $\{(\mathbf{x}_k, \vec{t}_k)\}_{k=1}^6$, this orthogonality criterion defines a continuous algebraic hypersurface $\det(\mathbf{A}) = 0$. Unlike the 3-node basis $\mathcal{L}^1$ where singular geometries admit straightforward spatial classifications (Fig. 2), the complete 6-node space $\mathcal{P}^1$ couples spatial coordinates and angular orientations into a rich manifold: $\mathbf{A}$ becomes singular whenever directors align with the local normals of an affine vector field, even on non-collinear point distributions with mutually non-parallel directors. Configurations in the immediate neighborhood of this hypersurface produce severely ill-conditioned matrices. Attempting to enumerate isolated geometric pathologies via heuristic decision trees is therefore intrinsically incomplete, establishing the necessity of qualifying candidate sextets directly through a unified algebraic metric.

3.3. Automated Support Selection Algorithm via Scale-Invariant Dimensionless Matrix

As established in Section 3.2, matrix singularity in the VNMM local placement system ($\mathbf{A}\boldsymbol{\beta}_i = \mathbf{L}_i$) is governed by the Null-Space Orthogonality Principle, forming an algebraic hypersurface in configuration space. Because singular and near-singular configurations arise whenever nodal directors approach pointwise orthogonality to an affine field in $\mathcal{P}^1$—regardless of whether the point distribution appears geometrically well-spread—heuristic geometric selection schemes relying on conditional decision trees are fundamentally incomplete and fragile.

In contrast, evaluating the magnitude of the determinant $|\det(\mathbf{A}_{6\times6})|$ of the local placement matrix serves as a unified algebraic filter against exact and near-singular configurations. Because $\mathbf{A}_{6\times6}$ incorporates both spatial coordinates and vector director projections within a single operator, enforcing $|\det(\mathbf{A}_{6\times6})| \geq \text{Tol}$ simultaneously screens out all configurations on the null-space orthogonality hypersurface as well as candidate sextets in their ill-conditioned near-singular vicinity.

3.3.1. Scale-Invariant Support Selection via Dimensionless Matrix

In local translated coordinates centered at the evaluation point $P(x_P, y_P)$, the first two columns of the classical placement matrix $\mathbf{A} \in \mathbb{R}^{6\times6}$ consist of dimensionless unit director components $t_{kx}, t_{ky} \sim \mathcal{O}(1)$, while columns 3–6 contain products of coordinates and directors, $\Delta x_k t_{kx}, \Delta y_k t_{kx}, \Delta x_k t_{ky}, \Delta y_k t_{ky} \sim \mathcal{O}(h)$ [m], where h denotes the local characteristic nodal spacing. Consequently, the standard placement matrix has mixed physical units ($[L]^0$ and $[L]^1$), and its determinant scales quartically:

$$\det(\mathbf{A}_{6\times6}) \sim \mathcal{O}(h^4). \tag{33}$$

Directly qualifying candidate stencils using the dimensional determinant introduces two severe difficulties:

1. **Scale Coupling and Threshold Decay:** Because $\det(\mathbf{A}) \propto h^4$, the determinant decays rapidly under spatial refinement ($h \to 0$), preventing the definition of a universal, scale-independent threshold across arbitrary domain sizes or multiscale meshes.
2. **Condition Number Inflation:** The mixed physical units create an artificial scaling imbalance between columns 1–2 ($\mathcal{O}(1)$) and columns 3–6 ($\mathcal{O}(h)$), inflating the 2-norm condition number as $\kappa_2(\mathbf{A}) \propto \mathcal{O}(1/h)$ as $h \to 0$ purely as a metric artifact.

To eliminate this $\mathcal{O}(h^4)$ scaling dependence and decouple algebraic qualification from physical dimensions, local spatial coordinates are normalized by a characteristic local length scale $h_{\text{local}} > 0$. For general point distributions, h_{local} is defined deterministically as the mean Euclidean distance to the first six nearest spatial neighbors retrieved from the k-d tree:

$$h_{\text{local}}(\mathbf{x}_P) = \frac{1}{6}\sum_{k=1}^{6} \|\mathbf{x}_{(k)} - \mathbf{x}_P\|, \tag{34}$$

or uniformly as $h_{\text{local}} = h_{\text{nom}}$ on quasi-uniform point clouds with nominal spacing h_{nom} (Section 5). Using normalized dimensionless local coordinates $\hat{x}_k = \Delta x_k / h_{\text{local}}$ and $\hat{y}_k = \Delta y_k / h_{\text{local}}$, row k of the dimensionless matrix $\widehat{\mathbf{A}} \in \mathbb{R}^{6\times 6}$ is assembled as:

$$\widehat{\mathbf{A}}_{k,:} = \begin{bmatrix} t_{kx} & t_{ky} & \hat{x}_k t_{kx} & \hat{y}_k t_{kx} & \hat{x}_k t_{ky} & \hat{y}_k t_{ky} \end{bmatrix}, \quad k = 1, \ldots, 6. \tag{35}$$

The dimensional matrix $\mathbf{A}$ and its dimensionless counterpart $\widehat{\mathbf{A}}$ satisfy the exact diagonal relation:

$$\mathbf{A} = \widehat{\mathbf{A}}\, \mathbf{D}_h, \quad \text{with } \mathbf{D}_h = \operatorname{diag}(1,\ 1,\ h_{\text{local}},\ h_{\text{local}},\ h_{\text{local}},\ h_{\text{local}}), \tag{36}$$

which establishes the determinant identity:

$$\det(\mathbf{A}) = h_{\text{local}}^4 \det(\widehat{\mathbf{A}}) \implies |\det(\widehat{\mathbf{A}})| = \frac{|\det(\mathbf{A})|}{h_{\text{local}}^4}. \tag{37}$$

Support qualification is thus formulated directly on the dimensionless determinant:

$$|\det(\widehat{\mathbf{A}})| \geq \widehat{\text{Tol}} > 0, \tag{38}$$

where $\widehat{\text{Tol}} \sim \mathcal{O}(1)$ is a universal threshold (e.g., $\widehat{\text{Tol}} \in [0.05, 1.50]$). Operating on $\widehat{\mathbf{A}}$ simultaneously resolves threshold decay and clears the artificial $\mathcal{O}(1/h)$ penalty from $\kappa_2(\widehat{\mathbf{A}})$, preserving compact nearest-neighbor stencils ($K_{\text{avg}} \approx 6.5$–$7.1$ on Cartesian lattices, Table 2, and 7.7–8.2 on irregular clouds, Section 5.7) without support dilation as $h \to 0$.

Any alternate length scale $h_{\text{local}} \to c\, h_{\text{local}}$ transforms the determinant quartically:

$$|\det(\widehat{\mathbf{A}})| \longrightarrow c^{-4} |\det(\widehat{\mathbf{A}})|. \tag{39}$$

Because candidate stencils are bounded within the immediate neighborhood ($\|\hat{\mathbf{x}}_k\| \leq C$), the spectral norm is bounded by the Frobenius norm ($\sigma_{\max}^2 \leq \|\widehat{\mathbf{A}}\|_F^2 \leq 6(1 + C^2)$) [32]. Since $|\det(\widehat{\mathbf{A}})| \leq \sigma_{\min}\sigma_{\max}^5$, enforcing $|\det(\widehat{\mathbf{A}})| \geq \widehat{\text{Tol}}$ guarantees $\sigma_{\min} \geq \widehat{\text{Tol}}/\sigma_{\max}^5 > 0$, effectively controlling conditioning ($\kappa_2(\widehat{\mathbf{A}}) \leq 5.9$ under 25% jitter; Section 5.3).

The vector shape functions and spatial derivatives at P are obtained

directly from $\widehat{\boldsymbol{\beta}} = \widehat{\mathbf{A}}^{-1}$ via $\boldsymbol{\beta} = \mathbf{D}_h^{-1}\widehat{\boldsymbol{\beta}}$:

$$\vec{N}_i(P) = \begin{bmatrix} \hat{\beta}_{1i} \\ \hat{\beta}_{2i} \end{bmatrix}, \tag{40}$$

$$(\nabla \times \vec{N}_i)_z = \frac{1}{h_{\text{local}}} \left(\hat{\beta}_{5i} - \hat{\beta}_{4i} \right), \tag{41}$$

$$\nabla \cdot \vec{N}_i = \frac{1}{h_{\text{local}}} \left(\hat{\beta}_{3i} + \hat{\beta}_{6i} \right), \tag{42}$$

factoring out the $\mathcal{O}(h^{-1})$ gradient scaling analytically through $1/h_{\text{local}}$.

3.3.2. The 4-Phase Selection Pipeline

To locate well-conditioned candidate sextets across unstructured point clouds without exhaustive $\mathcal{O}(N^2)$ evaluations, the selection framework couples orthogonal k-d tree spatial queries [33, 29, 34, 26, 35] with dimensionless algebraic qualification in a 4-phase pipeline:

1. **Phase 1: Spatial Neighborhood Query via k-d Tree:** An orthogonal k-d tree ($\mathcal{O}(N \log N)$ one-off construction) queries the $K_{\text{initial}} \geq 12$ nearest neighbors of $P(x_P, y_P)$ with average complexity $\mathcal{O}(\log N)$, defining h_{local} via Eq. (34).
2. **Phase 2: Proximity-Ordered Candidate Generation:** Candidate sextets are generated by fixing the node closest to P as an anchor and taking 5-combinations from subsequent nearest neighbors, ordered by increasing mean distance.
3. **Phase 3: Dimensionless Matrix Assembly:** Local coordinates are normalized ($\hat{x}_k = \Delta x_k / h_{\text{local}}, \hat{y}_k = \Delta y_k / h_{\text{local}}$) and $\widehat{\mathbf{A}}_{6\times6}$ is assembled via Eq. (35).
4. **Phase 4: Dimensionless Qualification with Early Stopping:** The first candidate sextet satisfying $|\det(\widehat{\mathbf{A}}_{6\times6})| \geq \widehat{\text{Tol}}$ (Eq. (38)) is immediately accepted, terminating the search loop.

3.3.3. Computational Complexity and Combinatorial Cost Analysis

While an initial neighborhood of $K_{\text{initial}} = 12$ nodes yields a theoretical worst-case bound of:

$$N_{\text{comb}}^{\max} = \binom{K_{\text{initial}}}{6} = \binom{12}{6} = 924 \text{ combinations}, \tag{43}$$

early stopping terminates the search as soon as an admissible support is found. As a result, the actual qualification count $\bar{N}_{\text{tested}}$ evaluates only a minimal fraction of these combinations. The total operational runtime across N_{eval} evaluation points is:

$$T_{\text{support}} = \underbrace{\mathcal{O}(N \log N)}_{k\text{-d tree construction}} + \sum_{p=1}^{N_{\text{eval}}} \left[\underbrace{\mathcal{O}(\log N)}_{\text{spatial query}} + \underbrace{\bar{N}_{\text{tested}}^{(p)} C_{\det(6\times 6)}}_{\text{observed algebraic qualification}} \right]. \tag{44}$$

Table 2 reports the empirical performance metrics measured across the six cavity refinement levels of Section 5.6 (3×3 Gauss quadrature, N_g scaling from 1,764 to 20,736, $\widehat{\text{Tol}} = 0.80$). Across all levels, the mean qualification count $\bar{N}_{\text{tested}}$ ranges from only 4.73 to 5.95 sextets (with maximum evaluations never exceeding 87). Support selection consumes merely 42–51 μs per Gauss point ($t_{\text{sup}} = 0.091$–0.881 s, representing 44%–49% of total assembly time t_{ass}). Even under 25% spatial jitter and random directors (Section 5.3), early stopping tests an average of $\bar{N}_{\text{tested}} \approx 48$–$120$ candidates (< 1 ms per evaluation point). Consequently, support selection is dominated by the logarithmic spatial neighbor query rather than combinatorial matrix evaluations.

Table 2: Computational cost of automated support selection across the six cavity discretization levels of Section 5.6 (3×3 Gauss-Legendre quadrature, $\widehat{\text{Tol}} = 0.80$, $K_{\text{initial}} = 12$).

Grid Mesh	**DoFs**	**Gauss Points**	**Mean Neighbors**	**Evaluated Sextets** (N_{tested})		**Support Time**	**Assembly Time**
$N_v \times N_v$	N_{DoF}	N_g	K_{avg}	**Mean** ($\bar{N}_{\text{tested}}$)	**Maximum**	t_{sup} [s]	t_{ass} [s]
15×15	169	1,764	6.56	**5.95**	87	0.091	0.187
22×22	400	3,969	6.53	**5.42**	87	0.186	0.414
29×29	729	7,056	6.53	**5.14**	87	0.321	0.717
36×36	1156	11,025	6.53	**4.96**	87	0.484	1.088
43×43	1681	15,876	6.49	**4.81**	87	0.683	1.529
49×49	2209	20,736	6.49	**4.73**	87	0.881	1.996

4. Variational Formulation of the Resonant Cavity Problem

To evaluate the spectral accuracy and verify the shifting of non-physical spurious gradient modes outside the operational bandwidth in the Vector Nodal Meshless Method (VNMM 2D), we formulate the electromagnetic vector eigenvalue problem in a two-dimensional resonant cavity $\Omega = [0, \pi] \times [0, \pi] \subset \mathbb{R}^2$ with perfect electric conductor (PEC) walls $\partial\Omega$ [2].

4.1. The Transverse Electric (TE_z) Vector Eigenvalue Problem

In a homogeneous, isotropic, and lossless dielectric medium ($\epsilon_r = 1.0, \mu_r = 1.0$), time-harmonic Transverse Electric modes (TE_z) are governed by the vector double-curl Helmholtz equation for the transverse electric field $\vec{E}(\mathbf{x}) = [E_x(x,y), E_y(x,y)]^T$ [3, 24]:

$$\nabla \times \left(\frac{1}{\mu_r} \nabla \times \vec{E} \right) = \lambda \, \epsilon_r \vec{E}, \quad \text{in } \Omega \tag{45}$$

$$\hat{n} \times \vec{E} = \mathbf{0}, \quad \text{on } \partial\Omega \tag{46}$$

where $\lambda = k_0^2 = \omega^2 \mu_0 \epsilon_0$ denotes the target wavenumber eigenvalue, ω is the angular resonance frequency, and $\hat{n}$ is the outward unit normal vector to boundary $\partial\Omega$. In two dimensions, the curl operator acting on a 2D vector field yields a scalar normal component $(\nabla \times \vec{E})_z = \partial E_y/\partial x - \partial E_x/\partial y$, while the vector curl of a scalar field u_z returns a vector in the transverse plane $\nabla \times u_z = [\partial u_z/\partial y, -\partial u_z/\partial x]^T$.

The exact analytical eigenvalues for the square PEC cavity are given by [2]:

$$\lambda_{m,n}^{\text{exact}} = m^2 + n^2, \quad m, n \in \{0, 1, 2, \dots\}, \quad (m + n > 0) \tag{47}$$

4.2. Ritz-Galerkin Weak Form with Div-Curl Regularization

To solve the vector boundary-value problem (45)–(46) numerically, the strong-form double-curl Helmholtz equation must be recast into a variational weak formulation. In standard electromagnetic field theory, testing the Helmholtz equation against an admissible vector field $\vec{W}$ and applying vector Green's first identity yields the canonical curl-curl weak form. Let the underlying Hilbert space of square-integrable vector fields with square-integrable curl satisfying homogeneous tangential PEC boundary conditions be defined as [3, 4, 7]:

$$\mathcal{S} \triangleq \left\{ \vec{E} \in H(\mathbf{curl}; \Omega) \;\middle|\; \hat{n} \times \vec{E} = \mathbf{0} \text{ on } \partial\Omega \right\}. \tag{48}$$

The classical, unregularized variational problem seeks eigenpairs $(\lambda, \vec{E}) \in \mathbb{R} \times \mathcal{S} \setminus \{\mathbf{0}\}$ satisfying:

$$\int_\Omega \frac{1}{\mu_r} (\nabla \times \vec{W})_z (\nabla \times \vec{E})_z \, d\Omega = \lambda \int_\Omega \epsilon_r \vec{W} \cdot \vec{E} \, d\Omega, \quad \forall \vec{W} \in \mathcal{S}. \tag{49}$$

While Eq. (49) is spectrally well-behaved when discretized using globally conforming Nédélec edge elements (where exact de Rham sequence preservation confines the infinite-dimensional static gradient nullspace strictly to $\lambda = 0$), implementing this standard formulation directly within a meshless framework triggers severe spectral contamination: non-physical, highly oscillatory spurious modes infiltrate the low-frequency physical spectrum.

4.2.1. Theoretical Origin: From Global Non-Conformity to Gradient Leakage

In standard Edge Finite Element Formulations (EFEM), the discrete space is globally $H(\mathbf{curl})$-conforming ($V_h \subset \mathcal{S}$) because shared tangential degrees of freedom across conforming element facets guarantee exact tangential trace continuity ($\hat{n} \times [\vec{E}^h] = \mathbf{0}$) [3, 4]. Under this global conformity, the discrete de Rham complex property $\mathrm{Im}(\nabla) \subset \ker(\nabla\times)$ holds exactly: irrotational gradient fields ($\vec{E} = \nabla\phi$) yield an identically zero curl ($\nabla \times (\nabla\phi) \equiv \mathbf{0}$) and map directly into the zero eigenvalue subspace ($\lambda = 0$), preventing spectrum contamination without auxiliary constraints [3, 6, 7].

In contrast, on unconstrained point clouds, the assembled meshless VNMM vector approximation is globally non-conforming:

$$V_h \not\subset \mathcal{S} \subset H(\mathbf{curl}; \Omega). \tag{50}$$

Although each local vector shape function $\vec{N}_i(\mathbf{x})$ possesses local $H(\mathbf{curl})$ regularity within its local support stencil (Section 2.3), active support stencils are selected dynamically across the domain via k-d tree spatial queries. Consequently:

1. **Non-Inheritance of Discrete de Rham Exactness:** Because tangential traces suffer minor non-polynomial jumps across shifting support boundaries, the discrete approximation does not inherit the exactness of the discrete de Rham sequence. Specifically, discrete gradient fields fail to preserve the irrotational property during quadrature-based evaluation:
$$\nabla_h \times \nabla_h \phi_h \neq \mathbf{0}. \tag{51}$$
2. **Discrete Gradient Nullspace Leakage:** Because $\nabla_h \times \nabla_h \phi_h \neq \mathbf{0}$ at background quadrature points, non-zero numerical curl residues leak the infinite-dimensional static gradient nullspace into the non-zero discrete spectrum, causing non-physical, highly oscillatory spurious modes to infiltrate the physical resonance band.

3. **The Methodological Chain:** This establishes the cohesive architectural chain of the proposed framework:

$$\boxed{\begin{array}{rl} \mathcal{L}^1 \text{ (derivative defect)} & \longrightarrow \mathcal{P}^1 \text{ (local completeness)} \\ & \Downarrow \\ \text{dynamic supports} & \longrightarrow V_h \not\subset H(\mathbf{curl}) \text{ (non-conformity)} \\ & \Downarrow \\ \text{loss of de Rham exactness} & \longrightarrow \text{gradient leakage (spurious modes)} \\ & \Downarrow \\ \mathbf{K}_{\text{div}} \text{ regularization} & \longrightarrow \text{spectral shifting of spurious modes} \end{array}} \tag{52}$$

This causal chain demonstrates that while local polynomial completeness ($\mathcal{P}^1$) is indispensable to resolve the local Jacobian and eliminate derivative aliasing, clearing the operational spectrum from non-conforming gradient leakage on unconstrained point clouds fundamentally requires shifting spurious modes through the divergence penalty operator developed next.

4.2.2. Div-Curl Regularized Weak Formulation

To shift these spurious gradient modes outside the operational bandwidth without introducing saddle-point system indefiniteness or auxiliary Lagrange multipliers, we adopt a **Ritz-Galerkin weak formulation with div-curl penalty regularization** [3, 4, 7]:

$$\int_\Omega \frac{1}{\mu_r}(\nabla \times \vec{W})_z(\nabla \times \vec{E})_z \, d\Omega + s_{\text{div}} \int_\Omega (\nabla \cdot \vec{W})(\nabla \cdot \vec{E}) \, d\Omega = \lambda \int_\Omega \epsilon_r \vec{W} \cdot \vec{E} \, d\Omega, \quad \forall \vec{W} \in \mathcal{S} \tag{53}$$

where $s_{\text{div}} > 0$ is the non-dimensional divergence penalty parameter [3].

Upon discretizing the electric field as $\vec{E}^h(\mathbf{x}) = \sum_{j=1}^M e_j \vec{N}_j(\mathbf{x})$ using the scalar projection DoFs $e_j = \vec{E}(\mathbf{x}_j) \cdot \vec{t}_j$ (Eq. (2)) and test functions as $\vec{W}^h(\mathbf{x}) = \sum_{i=1}^M d_i \vec{N}_i(\mathbf{x})$ with test coefficients $d_i = \vec{W}(\mathbf{x}_i) \cdot \vec{t}_i$, the second bilinear term in Eq. (53) defines the **divergence penalty matrix** $\mathbf{K}_{\text{div}} \in \mathbb{R}^{M \times M}$, whose entries $K_{\text{div},ij}$ are explicitly defined by:

$$K_{\text{div},ij} \triangleq \int_\Omega (\nabla \cdot \vec{N}_i)(\nabla \cdot \vec{N}_j) \, d\Omega \tag{54}$$

Consequently, the global stiffness matrix $\mathbf{K}$ in the discrete generalized eigenvalue problem $\mathbf{Ke} = \lambda\mathbf{Me}$ decomposes into the sum of the primary curl-curl stiffness matrix $\mathbf{K}_{\text{curl}}$ and the scaled divergence penalty matrix $s_{\text{div}}\mathbf{K}_{\text{div}}$:

$$\mathbf{K} = \mathbf{K}_{\text{curl}} + s_{\text{div}}\mathbf{K}_{\text{div}} \tag{55}$$

with individual entries of the primary curl-curl stiffness matrix defined by:

$$K_{\text{curl},ij} \triangleq \int_{\Omega} \frac{1}{\mu_r}(\nabla \times \vec{N}_i)_z(\nabla \times \vec{N}_j)_z \, d\Omega. \tag{56}$$

The interaction between divergence penalization, nullspace control, and physical field fidelity is structured into four complementary mechanisms [3, 4, 7]:

1. **Volumetric Regularization and Domain Topology:** The penalty term $s_{\text{div}} \int_{\Omega}(\nabla \cdot \vec{W})(\nabla \cdot \vec{E}) \, d\Omega$ provides elliptic regularization in $H(\mathbf{curl}) \cap H(\text{div})$ [3, 7]. In simply connected domains with homogeneous PEC walls, the static curl nullspace $\ker(\nabla\times) = \nabla H_0^1(\Omega)$ contains no non-trivial harmonic fields ($\Delta\phi = 0 \implies \phi \equiv 0$), ensuring that all static gradient fields have non-zero divergence ($\Delta\phi \neq 0$) and are penalizable (multiply connected domains or re-entrant corners would require cuts or weighted regularization [4, 36, 37, 38]).
2. **Selective Spectral Shifting via Divergence Energy:** Upon activating $s_{\text{div}}\mathbf{K}_{\text{div}}$, the discrete Rayleigh quotient expands as:

$$\lambda(\mathbf{e}) = \frac{\mathbf{e}^T\mathbf{K}_{\text{curl}}\mathbf{e}}{\mathbf{e}^T\mathbf{Me}} + s_{\text{div}}\frac{\mathbf{e}^T\mathbf{K}_{\text{div}}\mathbf{e}}{\mathbf{e}^T\mathbf{Me}}. \tag{57}$$

   Because non-physical gradient states have $\nabla \cdot (\nabla\phi) = \Delta\phi \neq 0$, their divergence energy shifts their eigenvalues upward proportional to s_{div}, clearing the low-frequency operational bandwidth (Sections 5.5.3 and 5.5.4).
3. **Metric for Solenoidal Field Integrity (R_{div}):** We define the discrete divergence-to-curl energy quotient:

$$R_{\text{div}} \triangleq \frac{\mathbf{e}^T\mathbf{K}_{\text{div}}\mathbf{e}}{\mathbf{e}^T\mathbf{K}_{\text{curl}}\mathbf{e}}. \tag{58}$$

   Physical solenoidal cavity modes satisfy $R_{\text{div}} \ll 1$, confirming solenoidal field fidelity, whereas non-physical irrotational gradient modes exhibit $R_{\text{div}} \gg 1$, providing an unambiguous discriminator for mode identification.

4. **Spectral Shifting vs. Volumetric Locking:** At the continuous level, solenoidal modes satisfy $\nabla \cdot \vec{E} \equiv 0$, so the penalty vanishes identically. On discrete meshless stencils, pointwise divergence is satisfied only to discretization order ($\nabla \cdot \vec{E}^h \sim \mathcal{O}(h) \neq 0$). Consequently, excessive penalization introduces volumetric stiffening (locking), slightly perturbing physical eigenvalues and elevating matrix conditioning. A calibrated operating window ($s_{\text{div}} \in [6.0, 10.0]$) achieves spectral clearance without locking (Section 5.5.4).

4.3. Integration Strategy: Cell-Based vs. Point-Based (EFG-Style)

Numerical evaluation of weak form integrals in Eq. (53) requires background cell quadrature. In early VNMM implementations, support nodes and placement matrices $\mathbf{A}$ were constructed once per cell at its centroid P_c [23, 24]. However, expanding shape functions across large cells ($dx > h$) induces severe Taylor truncation errors.

To obtain a mesh-connectivity-free field approximation, we employ a point-based Element-Free Galerkin (EFG) integration strategy [12, 29]:

1. **Background Integration Scaffolding:** A resolution-matched background Cartesian grid of $N_{cx} \times N_{cy} = (N_v - 1) \times (N_v - 1)$ quadrilateral cells ($dx = dy = h_{\text{nom}}$) covers Ω with 3×3 Gauss-Legendre quadrature ($N_g / N_{\text{DoF}} \approx 10$), avoiding under-integration artifacts (Section 5.5.5).
2. **Pointwise Support Selection (Section 3.3.2):** At each Gauss integration point P_g, an admissible support sextet $\{P_k, \vec{t}_k\}_{k=1}^{6}$ is determined on-the-fly via k-d tree queries and scale-invariant determinant qualification ($|\det(\widehat{\mathbf{A}}_{6\times6})| \geq \widehat{\text{Tol}}$).
3. **Pointwise Shape Function Evaluation (Section 3.3.1):** Anchoring coordinates directly at P_g ($\Delta x = 0, \Delta y = 0$), vector shape functions $\vec{N}_i(P_g)$ and spatial derivatives ($\nabla \times \vec{N}_i, \nabla \cdot \vec{N}_i$) are evaluated via Eqs. (40)–(42), with $1/h_{\text{local}}$ factored out analytically.

Following classical meshfree classification [12, 15, 10], the VNMM field approximation and shape functions are strictly meshless and connectivity-free; background cells serve solely as geometric integration scaffolding without constraining the interpolation topology.

4.4. Algebraic Discretization and Generalized Eigenvalue System

Under the discrete field representations $\vec{E}^h$ and $\vec{W}^h$ introduced in Section 4.2.2, substituting the vector shape functions into the regularized weak

form (53) yields the global generalized algebraic eigenvalue problem $\mathbf{Ke} = \lambda\mathbf{Me}$ [24, 26]. The global regularized stiffness matrix $\mathbf{K} = \mathbf{K}_{\text{curl}} + s_{\text{div}}\mathbf{K}_{\text{div}} \in \mathbb{R}^{M\times M}$ is assembled directly from the curl-curl and divergence penalty blocks defined in Eqs. (54)–(56), whereas the entries of the global mass matrix $\mathbf{M} \in \mathbb{R}^{M\times M}$ are governed by the permittivity-weighted bilinear product on the right-hand side of Eq. (53):

$$M_{ij} \triangleq \int_{\Omega} \epsilon_r \, \vec{N}_i \cdot \vec{N}_j \, d\Omega. \tag{59}$$

Homogeneous PEC boundary conditions are collocated directly at boundary nodes by setting $e_k = 0$ for all boundary nodes whose unit directors $\vec{t}_k$ align tangent to $\partial\Omega$ ($\vec{t}_k \parallel \partial\Omega$), which enforces $\vec{E}^h(\mathbf{x}_k) \cdot \vec{t}_k = 0$ pointwise [23, 26]. Eliminating these constrained degrees of freedom yields the reduced algebraic eigenvalue system:

$$\mathbf{K}_{\text{red}}\mathbf{e}_{\text{red}} = \lambda\mathbf{M}_{\text{red}}\mathbf{e}_{\text{red}}, \tag{60}$$

of size $M_{\text{red}} \times M_{\text{red}}$ (equivalently designated as $N_{\text{DoF}} \equiv M_{\text{red}}$ in subsequent convergence comparisons).

While the unregularized Maxwell stiffness operator is symmetric positive-semidefinite due to the static gradient nullspace, the inclusion of the divergence penalty ($s_{\text{div}} > 0$) combined with essential PEC boundary constraints renders $\mathbf{K}_{\text{red}}$ strictly positive-definite on the discrete physical subspace, while the mass matrix $\mathbf{M}_{\text{red}}$ is natively positive-definite. This ensures that the discrete generalized eigenvalue problem involves well-behaved, sparse symmetric positive-definite (SPD) reduced matrices, allowing efficient and stable computation of the physical spectrum using Krylov-subspace shift-and-invert Lanczos methods [39, 40], as implemented via the ARPACK solver (`eigsh`) in SciPy [41].

5. Numerical Verification and Validation

This section assesses the numerical performance of the proposed Vector Nodal Meshless Method over the canonical domain $\Omega = [0, \pi]^2$, organized into three complementary parts: (1) **Numerical Verification (Sections 5.1–5.4):** local dual equivalence against Whitney elements, restoration of asymptotic curl convergence via $\mathcal{P}^1$, algebraic stencil conditioning under spatial-directional jitter, and linear completeness and asymptotic convergence via

the Galerkin Patch Test and MMS; (2) **Physics-Based Spectral Validation (Sections 5.5–5.7):** resonant cavity spectra under div-curl penalty regularization, parametric sensitivity of s_{div}, comparative convergence benchmarking against Edge FEM, and spectral validation on irregular point clouds; and (3) **Methodological Scope and Boundaries (Section 5.8):** operational boundaries, background quadrature scaffolding, boundary stencil starvation along curved geometries, high-frequency dispersion, and extensions to 3D domains.

Baseline Computational Setup and Operational Conventions. To ensure methodological consistency and eliminate repetitive descriptions, Table 3 establishes the uniform baseline conventions and parameters adopted throughout Section 5. Because Section 5.1 evaluates $\mathcal{L}^1$ at triangular midpoints without dynamic selection, $\widehat{\mathbf{A}}$ and h_{local} are not invoked there.

5.1. Local Constructive Equivalence under Conforming Midpoint/Tangent Placement: VNMM $\mathcal{L}^1$ vs. Whitney EFEM

To establish a rigorous baseline connecting the proposed vector meshless formulation to established finite element theory, this first experiment numerically validates the Local Constructive Equivalence Theorem (Theorem 1) established in Section 2.2.3 and proved in Appendix A, comparing the local vector interpolation capacity of the incomplete 3-node basis $\mathcal{L}^1$ against lowest-order Whitney 1-form edge elements (Nédélec 1st-kind) [8, 3, 7].

5.1.1. Experimental Setup and Dual Topologies

The benchmark test considers the TE_{11} resonant mode over the square cavity domain $\Omega = [0, \pi] \times [0, \pi]$, governed by the exact analytical field [2, 1]:

$$\vec{E}_{\text{ex}}(x, y) = \begin{bmatrix} \cos(x)\sin(y) \\ -\sin(x)\cos(y) \end{bmatrix}, \qquad (\nabla \times \vec{E}_{\text{ex}})_z(x, y) = -2\cos(x)\cos(y) \quad (61)$$

To compare both formulations under identical spatial sampling, a sequence of seven conforming triangular meshes ($N \in \{4, 6, 8, 12, 16, 24, 32\}$ divisions per side) is generated with controlled stochastic vertex perturbation (10% jitter) to ensure arbitrary, non-equilateral simplices. All interpolation errors are evaluated across a fixed, invariant grid of $M = 1{,}600$ internal Cartesian points (40×40) distributed uniformly within $[0.04\pi, 0.96\pi]^2$. The characteristic mesh spacing is defined by the average edge length $h = h_{\text{avg}} = \frac{1}{N_e}\sum_{k=1}^{N_e} L_k$.

Table 3: Baseline computational setup, geometric discretizations, and operational solver conventions across Section 5.

Convention / Parameter	**Mathematical Specification / Operational Formulation**	**Applicable Benchmarks**
Domain & Interior Grid	$\Omega = [0,\pi]^2$, reference grid with $N_x \times N_y$ nodes ($N_x = N_y$), nominal spacing $h_{\text{nom}} = \pi/(N_x - 1)$. Cartesian interior coordinates: $\mathbf{x}_{i,j} = (i\,h_{\text{nom}},\ j\,h_{\text{nom}})$ for $i,j = 1,\dots,N_x - 2$.	All benchmarks without exception (Secs. 5.1–5.7)
Staggered Boundary Nodes	Midpoints along $\partial\Omega$ edges ($N_{\text{bnd}} = 4(N_x - 1)$ nodes): $y \in \{0,\pi\} : \mathbf{x}_k = ((k+\frac{1}{2})h_{\text{nom}}, y)$, $\vec{t}_k = [1,0]^T$; $\quad x \in \{0,\pi\} : \mathbf{x}_k = (x, (k+\frac{1}{2})h_{\text{nom}})$, $\vec{t}_k = [0,1]^T$. Distance $\geq h_{\text{nom}}/2$ from corners. Homogeneous PEC: $e_k = 0$.	Cavity Dirichlet and PEC benchmarks (Secs. 5.4–5.7)
Internal Directors (Structured)	Checkerboard alternating orthogonal unit orientations: $\vec{t}_{i,j} = [\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}]^T$ if $(i+j)$ is even; $\vec{t}_{i,j} = [-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}]^T$ if $(i+j)$ is odd. Guarantees local orthogonal diversity ($\vec{t}_{45^\circ} \cdot \vec{t}_{135^\circ} = 0$).	Regular lattice benchmarks (Secs. 5.4–5.6)
Stochastic Clouds (Disordered)	Interior nodes perturbed with 25% spatial jitter and uniform random director orientations: $\mathbf{x}_k = \mathbf{x}_k^{(0)} + (\delta x_k, \delta y_k)$, $\quad \delta x_k, \delta y_k \sim \mathcal{U}(-0.25\,h_{\text{nom}},\ 0.25\,h_{\text{nom}})$ $\vec{t}_k = [\cos\theta_k,\ \sin\theta_k]^T$, $\quad \theta_k \sim \mathcal{U}(0, 2\pi)$. Conforming unit tangents on $\partial\Omega$.	Disordered cloud benchmarks (Secs. 5.2, 5.3, 5.4, 5.7)
Support Qualification & Scaling	Scale normalization $h_{\text{local}} = h_{\text{nom}}$ (or metric (34)). Qualification threshold $\widehat{\text{Tol}} = 0.80$ with query pool $K_{\text{initial}} = 12$ nearest neighbors ($\lvert\det(\widehat{\mathbf{A}}_{6\times 6})\rvert \geq \widehat{\text{Tol}}$).	All $\mathcal{P}^1$ benchmarks (Secs. 5.2–5.7); swept in Sec. 5.3
Div-Curl Regularization	Penalty coefficient $s_{\text{div}} = 6.0$ in regularized weak form (53), shifting gradient nullspace modes out of the operational cavity spectrum while preserving physical accuracy.	All variational formulations (Secs. 5.4–5.7); swept in Sec. 5.5.4
Background Quadrature	Topologically independent Cartesian grid of $(N_x - 1) \times (N_y - 1)$ cells ($dx = dy = h_{\text{nom}}$) with 3×3 Gauss-Legendre quadrature (9 points/cell, $N_g/N_{\text{DoF}} \approx 10$).	All variational formulations (Secs. 5.4–5.7); swept in Sec. 5.5.5

The geometric arrangement follows the dual configuration established in Section 2.2.3 and illustrated in Fig. 3. For each triangular cell T, VNMM nodes are placed at the edge midpoints M_k with tangent directors $\vec{t}_k$, mapping discrete degrees of freedom according to the dual scaling law $\alpha_k = L_k e_k$ (where edge circulations α_k scale with edge lengths L_k to match nodal directional projections e_k, as established by the shape function relation $\vec{N}_k \equiv L_k \vec{w}_k$ in Eq. (16)).

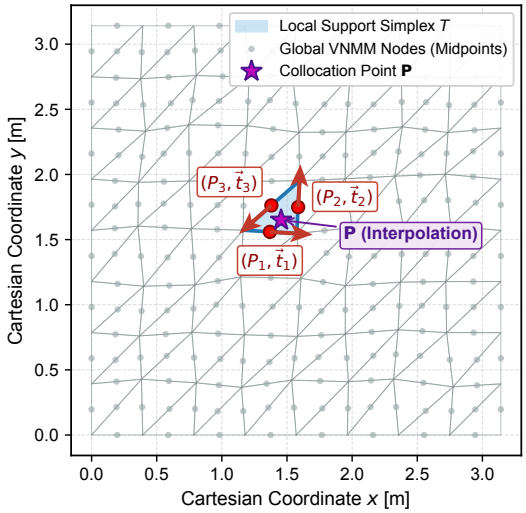


Figure 3: Conforming dual topology for Whitney EFEM and VNMM $\mathcal{L}^1$ over $\Omega = [0, \pi]^2$. Underlying triangular mesh overlaid with dual vector nodes placed at edge midpoints M_k with tangent directors $\vec{t}_k$.

5.1.2. Numerical Identity and Asymptotic Convergence Rates

The asymptotic convergence behavior comparing the VNMM 2D $\mathcal{L}^1$ and Whitney EFEM formulations across the seven conforming mesh levels is presented in Fig. 4.

As demonstrated in Fig. 4a, the electric field RMS error, defined over the M sampling points as

$$\|\vec{E} - \vec{E}^h\|_{\text{RMS}} = \sqrt{\frac{1}{M} \sum_{p=1}^{M} \|\vec{E}(\mathbf{x}_p) - \vec{E}^h(\mathbf{x}_p)\|_2^2}, \tag{62}$$

displays an identical first-order asymptotic decay rate of $\mathcal{O}(h^{1.00})$ for both formulations. The point-by-point field difference between the two methods remains strictly bounded by double-precision floating-point machine precision, satisfying $\|\vec{E}^{\mathrm{VNMM}} - \vec{E}^{\mathrm{EFEM}}\|_{\mathrm{RMS}} \leq 1.6 \times 10^{-16}$.

Similarly, Fig. 4b illustrates the curl operator RMS error $\|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z\|_{\mathrm{RMS}}$, which exhibits an identical fitted convergence rate of $\mathcal{O}(h^{0.96})$ for both formulations. The mutual curl field difference is likewise bounded by machine precision, satisfying $\|(\nabla \times \vec{E})^{\mathrm{VNMM}} - (\nabla \times \vec{E})^{\mathrm{EFEM}}\|_{\mathrm{RMS}} \leq 2.5 \times 10^{-15}$. These log-log convergence curves empirically confirm the theoretical equivalence established by Theorem 1 (Section 2.2.3 and Appendix A), establishing that under conforming midpoint/tangent placement on triangular cells, the VNMM 2D $\mathcal{L}^1$ formulation and Whitney edge finite elements reproduce identical discrete approximations down to double-precision floating-point roundoff.

It should be emphasized that this machine-precision equivalence holds strictly under conforming cell topologies where support triads match element midpoints with edge-tangent directors. On arbitrary point clouds, the VNMM ceases to be equivalent to a Whitney element, as arbitrary node coordinates, non-tangential director orientations, and dynamically shifting support stencils destroy simplex conformity, leading to the derivative stagnation examined next.

5.2. Vector Field Interpolation and Derivative Convergence in Arbitrary Meshless Clouds

To evaluate performance in practical meshless settings where nodes do not coincide with triangulation midpoints, we assess convergence across unconstrained point distributions. This validates the modal aliasing analysis of Section 2.2.5 and confirms its resolution through the complete linear space $\mathcal{P}^1$.

5.2.1. Experimental Setup on Unconstrained Point Clouds

The interpolation performance is assessed for the analytical TE_{11} cavity mode specified in Eq. (61) over unstructured, non-conforming nodal distributions covering $\Omega = [0, \pi] \times [0, \pi]$. Boundary nodes are placed uniformly along the domain perimeter with counterclockwise unit tangent vectors, whereas internal nodes are generated following the baseline stochastic model with 25% spatial jitter and uniform random director orientations (Table 3).

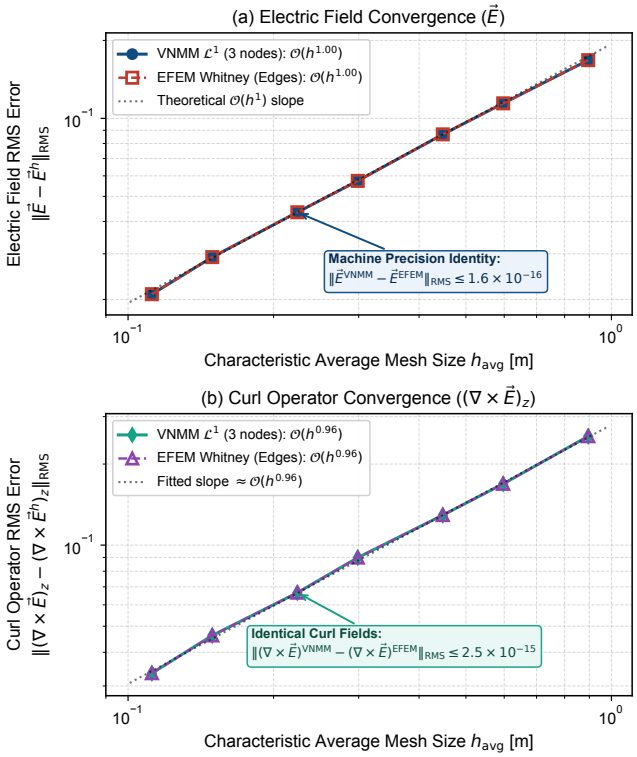


Figure 4: Convergence comparison between Whitney EFEM and VNMM $\mathcal{L}^1$ on conforming triangular meshes. (a) Electric field RMS error $\|\vec{E} - \vec{E}^h\|_{\text{RMS}}$ showing identical $\mathcal{O}(h^{1.00})$ convergence; (b) Curl operator RMS error displaying identical $\mathcal{O}(h^{0.96})$ convergence down to machine precision, confirming numerical identity.

To ensure spatial evaluation consistency, all field and derivative errors are evaluated across a fixed, invariant Cartesian grid of 1,600 internal sampling points (40×40) distributed uniformly in $[0.04\pi, 0.96\pi]^2$. The computational domain is discretized over six refinement levels, with total node counts $N_{\text{total}} \in \{84, 186, 416, 884, 1{,}928, 4{,}192\}$ and nominal unperturbed boundary spacings ranging from $h_{\text{nom}} = 4\pi/N_{\text{bnd}} = 0.5236$ m down to 0.0654 m ($h_{\text{nom}} \in \{0.52, 0.35, 0.22, 0.15, 0.10, 0.07\}$ m), where $N_{\text{bnd}} \in \{24, 36, 56, 84, 128, 192\}$ denotes the number of boundary nodes along the perimeter $\partial\Omega$ of length 4π.

5.2.2. Empirical Validation of Curl Stagnation in $\mathcal{L}^1$

When evaluated on unconstrained meshless clouds, the 3-node solenoidal basis $\mathcal{L}^1$ exhibits the fundamental breakdown in derivative convergence pre-

dicted theoretically in Section 2.2.5. As illustrated by the log-log curves in Fig. 5a, the electric field RMS error $\|\vec{E} - \vec{E}^h\|_{\text{RMS}}$ under $\mathcal{L}^1$ decreases monotonically at a first-order rate of $\mathcal{O}(h^{0.97})$, dropping from 1.91×10^{-1} ($h_{\text{nom}} = 0.5236$ m) down to 2.46×10^{-2} ($h_{\text{nom}} = 0.0654$ m).

However, as revealed in Fig. 5b, the RMS error of the curl operator $\|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z\|_{\text{RMS}}$ **stagnates at an $\mathcal{O}(1)$ error plateau** (≈ 0.85, varying between 8.33×10^{-1} and 8.88×10^{-1}, asymptotic rate $\mathcal{O}(h^{-0.01}) \approx \mathcal{O}(1)$), failing to converge under spatial refinement ($N_{\text{total}} = 4{,}192$). This persistent noise floor confirms the modal aliasing mechanism derived in Section 2.2.5, wherein unrepresented normal field derivatives leak into the rotational coefficient via the inverted placement matrix $\mathbf{A}^{-1}$.

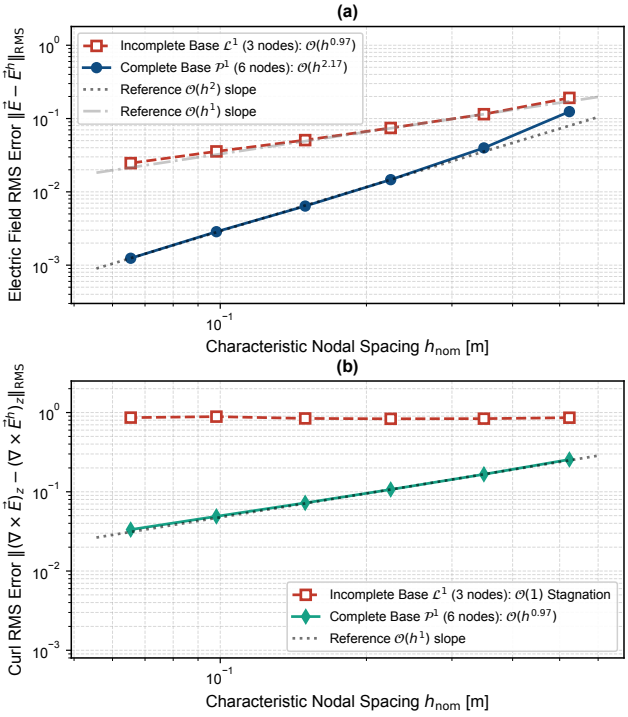


Figure 5: Convergence of $\mathcal{L}^1$ and $\mathcal{P}^1$ bases on unconstrained point clouds for the TE_{11} mode. (a) Electric field RMS error showing $\mathcal{O}(h^{0.97})$ ($\mathcal{L}^1$) versus $\mathcal{O}(h^{2.17})$ ($\mathcal{P}^1$); (b) Curl operator RMS error highlighting $\mathcal{O}(1)$ stagnation for $\mathcal{L}^1$ and restored $\mathcal{O}(h^{0.97})$ convergence for $\mathcal{P}^1$.

5.2.3. Restoration of Asymptotic Convergence via Complete $\mathcal{P}^1$ Basis

Expanding the local approximation space to the complete linear polynomial vector space $\mathcal{P}^1$ introduced in Section 2.3 provides six independent degrees of freedom, capturing all four components of the local Jacobian tensor $\nabla \vec{E}^h$. Active support sextets are selected via the baseline scale-invariant pipeline ($h_{\text{local}} = h_{\text{nom}}$, $\widehat{\text{Tol}} = 0.80$), yielding compact local stencils with $K_{\text{avg}} \approx 10.7$–$11.2$ across all six refinement levels.

As demonstrated in Fig. 5, transitioning to $\mathcal{P}^1$ removes derivative modal aliasing and achieves two major performance gains:

1. **Observed Convergence for Field $\vec{E}^h$ (Fig. 5a):** Completing the linear polynomial space elevates the observed electric field convergence rate from $\mathcal{O}(h^{0.97})$ under $\mathcal{L}^1$ to an observed fitted slope of $\mathcal{O}(h^{2.17})$ under $\mathcal{P}^1$. At the finest discretization ($h_{\text{nom}} = 0.0654$ m), the field RMS error drops from 2.46×10^{-2} ($\mathcal{L}^1$) down to 1.24×10^{-3} ($\mathcal{P}^1$), representing a $19.8\times$ **precision gain**. This empirical convergence acceleration highlights the substantial accuracy benefit of resolving the complete local Jacobian, in contrast to the incomplete $\mathcal{L}^1$ formulation whose missing longitudinal polynomial terms restrict error reduction.
2. **Restoration of Asymptotic $\mathcal{O}(h^1)$ Curl Convergence (Fig. 5b):** By accommodating normal derivatives in the Taylor expansion, the complete $\mathcal{P}^1$ basis restores first-order convergence $\mathcal{O}(h^{0.97})$ for the curl operator. The curl RMS error decreases monotonically from 2.55×10^{-1} down to 3.33×10^{-2} at $h_{\text{nom}} = 0.0654$ m, delivering a $25.9\times$ **error reduction** over the stagnated $\mathcal{L}^1$ formulation (8.63×10^{-1}).

Consequently, while $\mathcal{L}^1$ is suitable for conforming triangular meshes where circulation cancellation holds (Section 5.1), adopting the complete $\mathcal{P}^1$ vector space is sufficient to eliminate the derivative-aliasing mechanism and restore first-order curl convergence across unconstrained, arbitrary meshless point clouds. The quantitative performance comparison between the two bases across all six refinement levels is detailed in Table 4.

5.3. Numerical Robustness in the Tested Stochastic Realization Under Spatial-Directional Perturbations

While Section 5.2 adopted a fixed threshold ($\widehat{\text{Tol}} = 0.80$) to demonstrate curl convergence restoration via $\mathcal{P}^1$, this section investigates how $\widehat{\text{Tol}}$ governs stencil qualification, algebraic stability, and interpolation accuracy. On

Table 4: Comparative interpolation and curl error metrics between $\mathcal{L}^1$ and $\mathcal{P}^1$ bases on perturbed point clouds for the TE_{11} mode across six refinement levels.

N_{total}	h_{nom} [m]	**Electric Field RMS Error**			**Curl Operator RMS Error**			K_{avg}
		$\mathcal{L}^1$	$\mathcal{P}^1$	**Gain**	$\mathcal{L}^1$	$\mathcal{P}^1$	**Gain**	($\mathcal{P}^1$)
84	0.5236	1.91×10^{-1}	1.24×10^{-1}	**1.5**×	8.61×10^{-1}	2.55×10^{-1}	**3.4**×	11.1
186	0.3491	1.15×10^{-1}	3.97×10^{-2}	**2.9**×	8.38×10^{-1}	1.66×10^{-1}	**5.0**×	11.2
416	0.2244	7.44×10^{-2}	1.47×10^{-2}	**5.1**×	8.34×10^{-1}	1.07×10^{-1}	**7.8**×	11.2
884	0.1496	5.08×10^{-2}	6.39×10^{-3}	**7.9**×	8.43×10^{-1}	7.24×10^{-2}	**11.6**×	10.9
1,928	0.0982	3.56×10^{-2}	2.86×10^{-3}	**12.5**×	8.88×10^{-1}	4.89×10^{-2}	**18.1**×	10.9
4,192	0.0654	2.46×10^{-2}	1.24×10^{-3}	**19.8**×	8.63×10^{-1}	3.33×10^{-2}	**25.9**×	10.7
Fitted Rate		$\mathcal{O}(h^{0.97})$	$\mathcal{O}(\mathbf{h^{2.17}})$	—	$\mathcal{O}(h^{-0.01})$	$\mathcal{O}(\mathbf{h^{0.97}})$	—	—

unconstrained point clouds with spatial and directional disorder, arbitrary arrangements can approach the null-space orthogonality singularity condition (Section 3.2), degrading the conditioning of $\mathbf{A} \in \mathbb{R}^{6\times6}$. Using the baseline stochastic cloud sequence (Table 3) across $N_{\text{total}} = 84$ to 4,192, we evaluate the performance across $\widehat{\text{Tol}} \in [0.05, 6.00]$. Figure 6 and Table 5 summarize the resulting convergence, conditioning, and parametric sensitivity.

5.3.1. Degradation Under Permissive Thresholds vs. Scale-Invariant Stabilization

Under spatial-directional perturbations, an excessively relaxed threshold ($\widehat{\text{Tol}} \leq 0.05$) causes early-stopping qualification to accept immediate spatial neighbors ($K_{\text{avg}} \approx 6.3$). Because these candidate stencils frequently approach the null-space orthogonality hypersurface of Section 3.2, the condition number inflates to $\kappa_2(\widehat{\mathbf{A}}) > 13$ (unscaled $\kappa_2(\mathbf{A}) > 180$). As spatial derivatives scale as $\mathcal{O}(h^{-1})$ via Eq. (41), the ill-conditioned inverse $\widehat{\mathbf{A}}^{-1}$ amplifies local errors, causing the curl RMS error to flatten into an error plateau ($\approx 6.0 \times 10^{-2}$ to 8.7×10^{-2}, dashed crimson curve in Fig. 6a).

Enforcing a scale-invariant lower bound $\widehat{\text{Tol}} \geq 0.80$ completely eliminates this instability. Candidate sextets approaching the null-space orthogonality singularity are systematically rejected, moderately expanding the search radius (K_{avg} expands from 6.3 to 10.3) to identify well-conditioned sextets with balanced angular diversity. As shown by the solid curves in Fig. 6a, this restores monotonic convergence for both field $\vec{E}^h$ and curl $(\nabla \times \vec{E}^h)_z$, reducing fine-mesh curl error to 3.18×10^{-2} on $N_{\text{total}} = 4{,}192$. Because qualification operates on $\widehat{\mathbf{A}}$, the threshold $\widehat{\text{Tol}}$ remains scale-invariant across all refinement

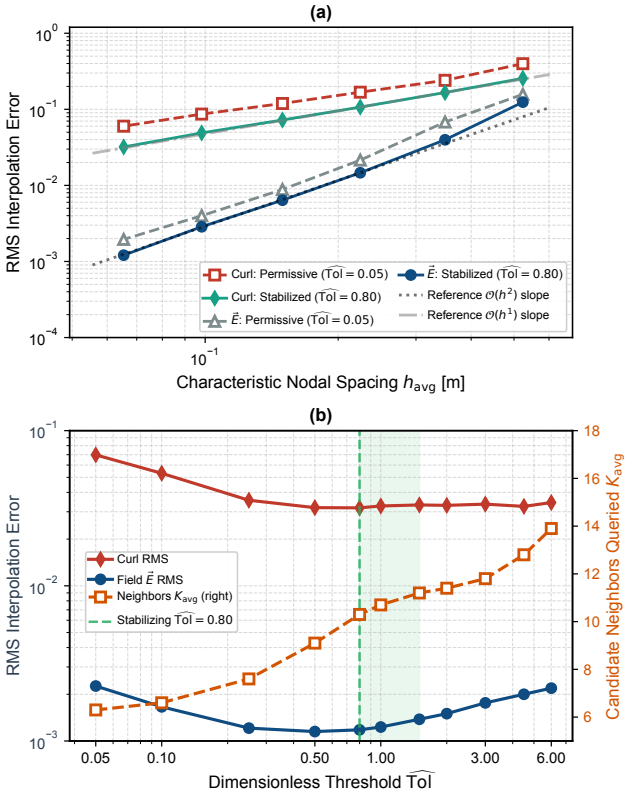


Figure 6: Conditioning and robustness of the VNMM $\mathcal{P}^1$ basis under spatial-directional perturbations. (a) Electric field and curl RMS errors versus nominal spacing h_{nom}, comparing a permissive threshold ($\widehat{\mathrm{Tol}} = 0.05$) against the calibrated threshold ($\widehat{\mathrm{Tol}} = 0.80$); (b) Error metrics and average support size K_{avg} versus $\widehat{\mathrm{Tol}}$ on the fine cloud ($N_{\mathrm{total}} = 4{,}192$), identifying the stable window $\widehat{\mathrm{Tol}} \in [0.80, 1.50]$.

levels, avoiding support dilation as $h \to 0$.

5.3.2. Parametric Trade-off and Stable Operating Window

To evaluate the sensitivity and identify a suitable operating regime, a parametric sweep was conducted on the fine cloud ($N_{\mathrm{total}} = 4{,}192$) for $\widehat{\mathrm{Tol}} \in [0.05, 6.00]$ (Table 5 and Fig. 6b). All 1,600 evaluation points successfully identified admissible supports without search starvation across all thresholds.

As $\widehat{\mathrm{Tol}}$ increases from 0.05 to 0.80, the curl RMS error drops monotonically by 2.2× ($6.98 \times 10^{-2} \to 3.18 \times 10^{-2}$), bounding conditioning to $\kappa_2(\widehat{\mathbf{A}}) \leq 5.9$ with compact local stencils ($K_{\mathrm{avg}} = 10.3$). In the window $\widehat{\mathrm{Tol}} \in$

$[0.80, 1.50]$, accuracy and conditioning remain stable ($K_{\text{avg}} \leq 11.2, \kappa_2 \leq 5.9$). For $\widehat{\text{Tol}} \geq 3.00$, conditioning improves marginally ($\kappa_2 \approx 3.3$–4.0) at the cost of broader support stencils ($K_{\text{avg}} \to 13.9$ at $\widehat{\text{Tol}} = 6.00$) and increased field interpolation errors (+85%). Thus, $\widehat{\text{Tol}} \in [0.80, 1.50]$ represents a well-balanced compromise between conditioning, locality, and precision.

Table 5: Sensitivity of error metrics, support neighborhood K_{avg}, and conditioning to the dimensionless threshold $\widehat{\text{Tol}}$ on the fine point cloud ($N_{\text{total}} = 4{,}192$).

Dimensionless Threshold	**Neighbors**	**RMS Error**		**Conditioning**
$\widehat{\text{Tol}}$	K_{avg}	$\vec{E}^h$	$(\nabla \times \vec{E}^h)_z$	$\kappa_2(\widehat{\mathbf{A}})$ $[\kappa_2(\mathbf{A})]$
0.05 (Permissive)	6.3	2.26×10^{-3}	6.98×10^{-2}	≈ 13.4 [186] (Ill-cond.)
0.10	6.6	1.66×10^{-3}	5.31×10^{-2}	≈ 11.8 [162]
0.25	7.6	1.21×10^{-3}	3.56×10^{-2}	≈ 9.0 [124]
0.50	9.1	1.15×10^{-3}	3.19×10^{-2}	≈ 6.9 [94]
0.80 (Stable)	**10.3**	$\mathbf{1.18 \times 10^{-3}}$	$\mathbf{3.18 \times 10^{-2}}$	$\approx$ **5.9** [**80**] **(Stable)**
1.00	10.7	1.23×10^{-3}	3.27×10^{-2}	≈ 5.6 [75]
1.50 (Compromise)	**11.2**	$\mathbf{1.38 \times 10^{-3}}$	$\mathbf{3.32 \times 10^{-2}}$	$\approx$ **5.1** [**66**] **(Stable)**
2.00	11.4	1.50×10^{-3}	3.30×10^{-2}	≈ 4.6 [59]
3.00	11.8	1.76×10^{-3}	3.36×10^{-2}	≈ 4.0 [50]
6.00 (Over-constrained)	13.9	2.19×10^{-3}	3.44×10^{-2}	≈ 3.3 [39]

5.4. Variational Verification: Galerkin Patch Test and Method of Manufactured Solutions

While Section 5.2 established that the complete linear polynomial vector space $\mathcal{P}^1$ restores $\mathcal{O}(h^1)$ convergence in pointwise collocation, true convergence and numerical stability of a Ritz-Galerkin meshless formulation further demand *linear completeness* and systematic asymptotic convergence in the presence of continuous differential operators [12, 10]. In this section, the discrete Galerkin operator is subjected to two complementary levels of numerical verification:

1. **The Galerkin Patch Test:** Verifies that the global discrete operator reproduces arbitrary affine polynomial fields (with spatially constant curl and divergence) exactly to machine precision without discretization bias under inhomogeneous boundary conditions.

2. **The Method of Manufactured Solutions (MMS):** Verifies the asymptotic convergence rates $\mathcal{O}(h^p)$ under spatial refinement ($h \to 0$) for a driven Maxwell/Helmholtz boundary value problem driven by a distributed volumetric source and spatially varying curl.

5.4.1. Part A: Galerkin Patch Test for Affine Vector Fields

Consider an arbitrary affine 2D vector field:

$$\vec{E}^*(\mathbf{x}) = \begin{bmatrix} a_1 + b_1 x + b_2 y \\ a_2 + b_3 x + b_4 y \end{bmatrix}, \quad a_i, b_i \in \mathbb{R}. \tag{63}$$

Because all first-order spatial derivatives of $\vec{E}^*$ are spatially constant across Ω, its curl and divergence evaluate to invariant scalars:

$$(\nabla \times \vec{E}^*)_z = b_3 - b_2 = C_{\text{curl}}, \qquad \nabla \cdot \vec{E}^* = b_1 + b_4 = C_{\text{div}}. \tag{64}$$

Consequently, the physical vector PDE $\nabla \times \nabla \times \vec{E}^* - s_{\text{div}} \nabla(\nabla \cdot \vec{E}^*) = \mathbf{0}$ is satisfied identically in Ω. In the regularized Ritz-Galerkin weak form (53), the continuous bilinear form evaluated against any admissible test function $\vec{W} = \vec{N}_i$ yields the discrete load component:

$$F_i \triangleq \int_\Omega \left[\frac{1}{\mu_r} (\nabla \times \vec{N}_i)_z C_{\text{curl}} + s_{\text{div}} (\nabla \cdot \vec{N}_i) C_{\text{div}} \right] d\Omega. \tag{65}$$

For any *harmonic affine field* ($C_{\text{curl}} = 0$ and $C_{\text{div}} = 0$), $F_i \equiv 0$ identically, reducing the patch test to a pure homogeneous interior Dirichlet problem without external forcing.

To span the full 6-dimensional linear vector space $\mathcal{P}_1 \times \mathcal{P}_1$, we test seven canonical affine field configurations: (1) **Translation X:** $\vec{E}^* = [1, 0]^T$ (harmonic); (2) **Translation Y:** $\vec{E}^* = [0, 1]^T$ (harmonic); (3) **Normal Strain:** $\vec{E}^* = [x, -y]^T$ (harmonic); (4) **Shear Strain:** $\vec{E}^* = [y, x]^T$ (harmonic); (5) **Pure Curl:** $\vec{E}^* = [-y, x]^T$ ($C_{\text{curl}} = 2, C_{\text{div}} = 0$); (6) **Pure Divergence:** $\vec{E}^* = [x, y]^T$ ($C_{\text{curl}} = 0, C_{\text{div}} = 2$); and (7) **General Affine Field:** $\vec{E}^* = [1.5 + 0.8x - 0.3y,\ -0.7 + 0.4x + 1.2y]^T$ ($C_{\text{curl}} = 0.7, C_{\text{div}} = 2.0$).

The domain $\Omega = [0, \pi]^2$ is discretized using $N_x = N_y = 11$ ($N_{\text{total}} = 121$ nodes, nominal spacing $h_{\text{nom}} = \pi/10 \approx 0.3142$ m). Boundary nodes ($N_{\text{bnd}} = 40$) are placed along $\partial\Omega$ using the baseline staggered face-centered arrangement (Table 3), where exact Dirichlet degrees of freedom are imposed:

$$e_k = \vec{E}^*(\mathbf{x}_k) \cdot \vec{t}_k, \quad \forall k \in \text{boundary nodes}. \tag{66}$$

The global linear system is partitioned into free interior degrees of freedom $\mathbf{e}_{\text{int}} \in \mathbb{R}^{81}$ and prescribed boundary degrees of freedom $\mathbf{e}_{\text{bnd}} \in \mathbb{R}^{40}$:

$$\begin{bmatrix} \mathbf{K}_{\text{int,int}} & \mathbf{K}_{\text{int,bnd}} \\ \mathbf{K}_{\text{bnd,int}} & \mathbf{K}_{\text{bnd,bnd}} \end{bmatrix} \begin{bmatrix} \mathbf{e}_{\text{int}} \\ \mathbf{e}_{\text{bnd}} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_{\text{int}} \\ \mathbf{F}_{\text{bnd}} \end{bmatrix} \implies \mathbf{K}_{\text{int,int}}\mathbf{e}_{\text{int}} = \mathbf{F}_{\text{int}} - \mathbf{K}_{\text{int,bnd}}\mathbf{e}_{\text{bnd}}. \tag{67}$$

To probe sensitivity to spatial irregularity, the patch test is solved on two contrasting nodal configurations: (i) a structured Cartesian grid with alternating internal directors; and (ii) an irregular cloud generated under the baseline stochastic model (both specified in Table 3).

Table 6: Galerkin patch test results: relative error in free interior degrees of freedom ($\|\mathbf{e}_{\text{int}} - \mathbf{e}^*_{\text{int}}\|_\infty / \|\mathbf{e}^*_{\text{int}}\|_\infty$) comparing the complete $\mathcal{P}^1$ and incomplete $\mathcal{L}^1$ bases on structured and irregular point clouds (121 nodes, 81 free interior DoFs, $s_{\text{div}} = 6.0$).

Affine Field Mode	**Regular Lattice**		**Irregular Cloud (25% Jitter)**	
	$\mathcal{P}^1$ Complete	$\mathcal{L}^1$ Incomplete	$\mathcal{P}^1$ Complete	$\mathcal{L}^1$ Incomplete
1. Translation X: $\vec{E} = [1, 0]^T$	4.49×10^{-14}	3.93×10^{-15}	5.14×10^{-14}	1.34×10^{-12}
2. Translation Y: $\vec{E} = [0, 1]^T$	4.52×10^{-14}	3.45×10^{-15}	7.38×10^{-14}	1.42×10^{-12}
3. Normal Strain: $\vec{E} = [x, -y]^T$	2.14×10^{-14}	2.14×10^{-1}	5.07×10^{-14}	3.27×10^{-1}
4. Shear Strain: $\vec{E} = [y, x]^T$	2.78×10^{-14}	1.51×10^{-1}	7.28×10^{-14}	4.69×10^{-1}
5. Pure Curl (Rotation): $\vec{E} = [-y, x]^T$	1.92×10^{-14}	2.23×10^{-15}	1.25×10^{-14}	5.01×10^{-13}
6. Pure Divergence: $\vec{E} = [x, y]^T$	2.48×10^{-14}	7.91×10^{-2}	5.45×10^{-14}	3.84×10^{-1}
7. Arbitrary General Affine Field	2.71×10^{-14}	8.64×10^{-2}	5.25×10^{-14}	3.78×10^{-1}

Following standard patch-test conventions in meshless methods [12, 10], Table 6 compiles the discrete maximum nodal degree-of-freedom error ($\|\mathbf{e}_{\text{int}} - \mathbf{e}^*_{\text{int}}\|_\infty / \|\mathbf{e}^*_{\text{int}}\|_\infty$) to verify that no individual interior node violates exact linear completeness. Inspection of the numerical results reveals two fundamental conclusions:

1. **Linear Completeness and Patch-Test Satisfaction of $\mathcal{P}^1$:** For all seven affine modes on both regular and irregular point clouds, the complete linear vector basis $\mathcal{P}^1$ reproduces the exact interior degrees of freedom to double-precision machine precision ($\|\mathbf{e}_{\text{int}} - \mathbf{e}^*_{\text{int}}\|_\infty / \|\mathbf{e}^*_{\text{int}}\|_\infty \leq 7.38 \times 10^{-14}$). Furthermore, to verify continuous domain-wide fidelity beyond discrete nodes, evaluating the reconstructed solution across the background integration domain under 25% spatial jitter and random directors confirms exact field and derivative reproduction in the continuous L^2 norm: the relative field error satisfies $\|\vec{E}^h - \vec{E}^*\|_{L^2} / \|\vec{E}^*\|_{L^2} \leq 5.93 \times 10^{-14}$, the curl error satisfies $\|(\nabla \times \vec{E}^h)_z - C_{\text{curl}}\|_{L^2} \leq 1.92 \times 10^{-13}$,

and the divergence error satisfies $\|\nabla \cdot \vec{E}^h - C_{\text{div}}\|_{L^2} \leq 8.37 \times 10^{-14}$. These results provide numerical verification that the complete basis $\mathcal{P}^1$, in conjunction with scale-invariant stencil qualification and regularized Ritz-Galerkin assembly, satisfies the Galerkin patch test to machine precision across all tested affine states.

2. **Catastrophic Variational Breakdown of the Incomplete $\mathcal{L}^1$ Basis:** In contrast, the incomplete 3-node basis $\mathcal{L}^1$ passes the patch test only for rigid-body motions (Translations 1–2 and Rotation 5), which reside within its restricted solenoidal subspace. Notably, even for solenoidal and irrotational affine fields ($C_{\text{curl}} = 0, C_{\text{div}} = 0$), such as normal strain (Mode 3) and shear strain (Mode 4), $\mathcal{L}^1$ fails catastrophically, exhibiting substantial relative errors of 15.1%–21.4% on regular lattices and 32.7%–46.9% under spatial jitter. For the general affine field (Mode 7), $\mathcal{L}^1$ incurs an interior DoF error of 37.8%.

This failure directly corroborates the theoretical analysis of Heuristic 1: on unconstrained point clouds lacking conforming closed-contour cancellation, the inability of $\mathcal{L}^1$ to represent normal and shear Jacobian gradients corrupts not only localized interpolation, but fundamentally contaminates the assembled Galerkin stiffness operator $\mathbf{K}_{\text{curl}}$. Furthermore, because the $\mathcal{L}^1$ basis is strictly solenoidal ($\nabla \cdot \vec{N}_i \equiv 0$), the divergence penalty operator vanishes identically ($\mathbf{K}_{\text{div}} \equiv \mathbf{0}$), precluding any variational div-curl regularization on this incomplete space.

5.4.2. Part B: Driven Source Problem and Method of Manufactured Solutions (MMS)

While passing the patch test proves that the discrete Galerkin operator introduces no bias when the exact solution is affine, it does not probe the asymptotic rate of convergence when the solution possesses non-vanishing spatial curvature and inhomogeneous curl. To investigate the convergence rate of the Galerkin formulation under spatial refinement ($h \to 0$), we formulate a source-driven boundary value problem using the **Method of Manufactured Solutions** (MMS).

Consider the regularized vector Helmholtz equation posed on $\Omega = [0, \pi]^2$:

$$\nabla \times \nabla \times \vec{E} - s_{\text{div}} \nabla(\nabla \cdot \vec{E}) + \kappa \vec{E} = \vec{f}(\mathbf{x}) \quad \text{in } \Omega, \tag{68}$$

subject to inhomogeneous essential Dirichlet conditions $\vec{E} \cdot \vec{t} = \vec{E}^* \cdot \vec{t}$ on $\partial\Omega$, with reaction parameter $\kappa = 1.0$ and divergence penalty parameter $s_{\text{div}} = 6.0$.

We prescribe the analytical solenoidal target field:

$$\vec{E}^*(x,y) = \begin{bmatrix} \sin(x)\cos(y) \\ -\cos(x)\sin(y) \end{bmatrix}. \tag{69}$$

Direct analytical differentiation verifies that:

$$\nabla \cdot \vec{E}^* \equiv 0, \qquad (\nabla \times \vec{E}^*)_z = 2\sin(x)\sin(y), \tag{70}$$

$$\nabla \times (\nabla \times \vec{E}^*) = 2\vec{E}^*. \tag{71}$$

By construction, on all four boundary segments of $\partial\Omega$ ($x \in \{0,\pi\}$ and $y \in \{0,\pi\}$), $\sin(x) = 0$ or $\sin(y) = 0$, implying that $(\nabla \times \vec{E}^*)_z = 0$ identically on $\partial\Omega$. Therefore, the natural boundary flux integral $\int_{\partial\Omega}(\vec{n} \times (\nabla \times \vec{E})) \cdot \vec{W}\, ds$ in the integration by parts vanishes identically. Substituting $\vec{E}^*$ into Eq. (68) yields the exact volumetric source distribution:

$$\vec{f}(\mathbf{x}) = (2+\kappa)\vec{E}^*(\mathbf{x}) = 3\vec{E}^*(\mathbf{x}). \tag{72}$$

In the Ritz-Galerkin discretization, the discrete system takes the form:

$$\mathbf{A}\mathbf{e} = \mathbf{F}, \quad \text{with } \mathbf{A} = \mathbf{K}_{\text{curl}} + s_{\text{div}}\mathbf{K}_{\text{div}} + \kappa\mathbf{M}, \tag{73}$$

where $\mathbf{M}$ is the consistent vector mass matrix (Eq. (59) with $\epsilon_r = 1$), $\mathbf{K}_{\text{curl}}$ and $\mathbf{K}_{\text{div}}$ are the stiffness blocks defined in Eqs. (54)–(56), and the load vector entries are defined by $F_i = \int_\Omega \vec{N}_i \cdot \vec{f}(\mathbf{x})\, d\Omega$. Prescribing Dirichlet boundary degrees of freedom $e_k = \vec{E}^*(\mathbf{x}_k) \cdot \vec{t}_k$ on $\partial\Omega$, the interior degrees of freedom are obtained from $\mathbf{A}_{\text{int,int}}\mathbf{e}_{\text{int}} = \mathbf{F}_{\text{int}} - \mathbf{A}_{\text{int,bnd}}\mathbf{e}_{\text{bnd}}$. To quantify accuracy and asymptotic convergence under spatial refinement, the discrete solution is evaluated against the analytical field (69) using three relative error metrics: the relative L^2 field error $\varepsilon_{L^2}(\vec{E}) = \|\vec{E}^h - \vec{E}^*\|_{L^2}/\|\vec{E}^*\|_{L^2}$, the relative curl semi-norm error $\varepsilon_{\text{curl}}(\vec{E}) = \|(\nabla \times \vec{E}^h)_z - (\nabla \times \vec{E}^*)_z\|_{L^2}/\|(\nabla \times \vec{E}^*)_z\|_{L^2}$, and the maximum interior degree-of-freedom error $\varepsilon_{\text{DoF}} = \|\mathbf{e}_{\text{int}} - \mathbf{e}^*_{\text{int}}\|_\infty/\|\mathbf{e}^*_{\text{int}}\|_\infty$, while $\|\nabla \cdot \vec{E}^h\|_{L^2}$ tracks the domain divergence residual.

We test a progressive refinement sequence of six regular lattices with $N \in \{9, 13, 17, 21, 25, 29\}$, spanning nominal nodal spacings from $h_{\text{nom}} = 0.3927$ m down to 0.1122 m (81 to 841 degrees of freedom, with scale normalization $h_{\text{local}} = h_{\text{nom}}$). The numerical results are summarized in Table 7. Two decisive phenomena emerge from this MMS convergence study:

Table 7: Convergence of the Galerkin Method of Manufactured Solutions (MMS) under spatial refinement: relative L^2 field error, relative $H(\mathbf{curl})$ semi-norm error, maximum DoF error, and divergence residual comparing the complete $\mathcal{P}^1$ and incomplete $\mathcal{L}^1$ bases ($s_{\text{div}} = 6.0$, $\kappa = 1.0$, 3×3 Gauss points per cell).

Mesh Grid	**DoFs**	h_{nom} [m]	**Complete Basis $\mathcal{P}^1$**				**Incomplete $\mathcal{L}^1$**
$N \times N$	N_{total}		$\varepsilon_{L^2}(\vec{E})$	$\varepsilon_{\text{curl}}(\vec{E})$	ε_{DoF}	$\|\nabla \cdot \vec{E}^h\|_{L^2}$	$\varepsilon_{L^2}(\vec{E})$
9×9	81	0.3927	12.32%	29.05%	23.74%	4.24×10^{-1}	24.80%
13×13	169	0.2618	3.78%	21.23%	9.03%	2.07×10^{-1}	24.15%
17×17	289	0.1963	1.85%	14.75%	5.70%	1.29×10^{-1}	22.81%
21×21	441	0.1571	1.35%	12.50%	3.94%	1.23×10^{-1}	25.91%
25×25	625	0.1309	1.34%	17.47%	4.75%	1.11×10^{-1}	23.60%
29×29	841	0.1122	0.77%	9.39%	2.61%	7.58×10^{-2}	23.87%
Empirical Convergence Rate $\mathcal{O}(h_{\text{nom}}^p)$:			**$\mathbf{p = 2.08}$**	**$\mathbf{p = 0.75}$**	**$\mathbf{p = 1.60}$**		**$\mathbf{p = 0.02}$**

1. **Systematic Asymptotic Convergence for $\mathcal{P}^1$:** As the nodal spacing is refined, the complete $\mathcal{P}^1$ basis exhibits strict monotonic convergence in the L^2 field norm, reducing the relative error from 12.32% on the coarsest mesh down to 0.77% on $N = 29 \times 29$. A least-squares logarithmic regression yields an empirical convergence rate of $p = 2.08$ in the L^2 field norm, in close agreement with the asymptotic $\mathcal{O}(h_{\text{nom}}^2)$ rate characteristic of linear approximations in L^2. Simultaneously, the $H(\mathbf{curl})$ semi-norm error converges at an empirical rate of $p = 0.75$, dropping from 29.05% to 9.39%, while the discrete degree-of-freedom error decays as $\mathcal{O}(h_{\text{nom}}^{1.60})$. The sub-unitary slope in $H(\mathbf{curl})$ reflects pre-asymptotic interaction between dynamic support transitions and the divergence penalty operator. The divergence residual also contracts by nearly an order of magnitude, confirming that the divergence penalty successfully restrains divergence errors.
2. **Convergence Stagnation of the Incomplete $\mathcal{L}^1$ Basis:** In stark contrast, the incomplete $\mathcal{L}^1$ basis fails to converge under refinement, exhibiting an asymptotic rate of $p = 0.02$. The relative L^2 field error remains trapped between 22.8% and 25.9%, while the curl error hovers near 55%–61%. Because the 3-node stencil $\mathcal{L}^1$ cannot represent normal derivative gradients on arbitrary node distributions, the discrete Galerkin stiffness operator cannot capture smooth vector field variations, leading to systematic stagnation regardless of grid refinement.

Together with the patch test, these MMS results provide systematic numerical evidence that adopting the complete linear polynomial vector space $\mathcal{P}^1$

resolves the variational breakdown and convergence stagnation of the incomplete $\mathcal{L}^1$ basis, restoring systematic asymptotic convergence in Galerkin meshless discretizations.

5.5. Spectral Validation: Resonant Cavity Analysis and Div-Curl Regularization

Having established the variational framework for the electromagnetic vector eigenvalue problem in Section 4, this experiment evaluates the spectral accuracy of the Vector Nodal Meshless Method and verifies the shifting of non-physical spurious modes outside the operational frequency band. Numerical simulations are conducted for Transverse Electric (TE_z) modes over the square PEC cavity using the complete $\mathcal{P}^1$ basis, comparing discrete spectra against the analytical reference eigenvalues defined in Eq. (47).

Numerical evaluations are conducted on the base-case discretization with $N_x = N_y = 21$ ($N_{\text{total}} = 441$ nodes, $N_{\text{bnd}} = 80$, $N_{\text{DoF}} = 361$ active internal degrees of freedom, $h_{\text{nom}} = \pi/20 \approx 0.1571$ m) adhering to the baseline operational conventions established in Table 3: regular Cartesian coordinates, staggered face-centered boundary placement, checkerboard alternating internal directors, and a background Cartesian grid of 20×20 cells with 3×3 Gauss quadrature ($N_g = 3{,}600$, $N_g/N_{\text{DoF}} \approx 10.0$).

Under the baseline qualification threshold ($\widehat{\text{Tol}} = 0.80$), the geometric symmetry and orthogonal angular diversity of immediate neighbors generate well-conditioned placement matrices across all 3,600 quadrature points (mean $|\det(\widehat{\mathbf{A}})| \approx 3.29$), yielding compact nearest-neighbor stencils ($K_{\text{avg}} \approx 6.54$) that remain identically invariant for any $\widehat{\text{Tol}} \leq 1.00$. Homogeneous tangential PEC boundary conditions ($\vec{E}^h(\mathbf{x}_k) \cdot \vec{t}_k = 0$) are enforced via nodal elimination ($e_k = 0$).

5.5.1. Spectral Contamination in the Unregularized Curl-Curl Formulation ($s_{\text{div}} = 0.0$)

As established in Section 4.2, meshless vector discretizations inherently lack discrete de Rham complex exactness due to global non-conformity ($V_h \not\subset H(\mathbf{curl})$), allowing discrete gradient fields to produce non-vanishing numerical curl residues ($\nabla_h \times \nabla_h \phi_h \neq \mathbf{0}$) at Gauss quadrature points. When solving the unregularized curl-curl formulation ($s_{\text{div}} = 0.0$) on the base-case cavity ($N_x = N_y = 21$), this absence of an exact sequence triggers catastrophic spectral contamination.

One might suspect that such spurious modes stem merely from numerical under-integration. However, evaluating the discrete operator $\mathbf{K}_{\text{curl}}$ under the dense baseline quadrature (20×20 cells, 3×3 Gauss points, $N_g/N_{\text{DoF}} \approx 10.0$) rules out this hypothesis: the operator exhibits an empty discrete nullspace ($\dim \ker(\mathbf{K}_{\text{curl}}) = 0$), proving that the Gauss points fully sample the discrete gradient field. Yet, because their numerical curl residues do not vanish, these unconstrained discrete gradient states leak directly into the positive spectrum, generating a dense ladder of non-physical spurious eigenvalues $\lambda \in (0.001, 0.90)$ strictly below the fundamental physical resonance ($TE_{10}, \lambda = 1.0$).

The empirical impact of this nullspace leakage is illustrated in Fig. 7a. Without volumetric divergence regularization ($s_{\text{div}} = 0.0$), over 130 non-physical gradient modes densely infiltrate the sub-resonance and operational spectrum ($\lambda \leq 10$), completely corrupting analytical modal ordering and yielding massive cutoff wavenumber errors in the lowest discrete modes (96.57% in Table 9).

5.5.2. Spectral Purification via Div-Curl Penalty Regularization ($s_{\text{div}} = 6.0$)

To shift the non-physical spurious gradient modes identified in Section 5.5.1 outside the operational band without introducing auxiliary Lagrange multipliers or expanding the system matrix, the div-curl variational penalty parameter is activated with $s_{\text{div}} = 6.0$ in Eq. (53). In accordance with the selective mechanism established in Section 4.2, this penalty selectively stiffens non-physical gradient states, shifting them past the 10th physical cavity mode ($\lambda_{\text{spur}} = 11.84 > 10.37$), thereby clearing the operational spectrum of interest ($\lambda \leq 10$) of detected non-physical states, while leaving solenoidal physical modes practically unaffected ($R_{\text{div}} \leq 1.65\%$). As analyzed in Section 5.5.4, larger penalty parameters (such as $s_{\text{div}} = 50.0$) can further elevate spurious modes to $\lambda_{\text{spur}} > 50$, though at the expense of increased matrix conditioning and locking tendencies.

The experimental confirmation of this selective purification is demonstrated in Fig. 7b. Activating $s_{\text{div}} = 6.0$ ensures that no spurious modes are detected among the first ten computed resonances within the tested operational spectral window ($\lambda \leq 10$), removing non-physical gradient branches from the low-frequency discrete spectrum and restoring 1-to-1 modal correspondence with the analytical reference eigenvalues defined in Eq. (47).

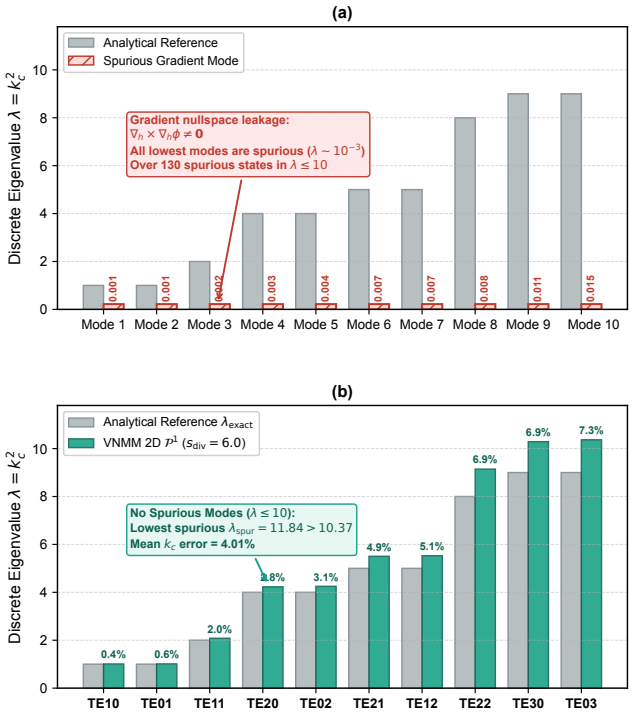


Figure 7: Discrete eigenvalue spectrum $\lambda = k_c^2$ for the TE_z square PEC cavity. (a) Unregularized curl-curl formulation ($s_{\text{div}} = 0.0$), showing spectral contamination from spurious gradient modes; (b) Div-curl regularized formulation ($s_{\text{div}} = 6.0$), displaying a purified spectrum matching analytical modes within the operational bandwidth ($\lambda \leq 10$).

5.5.3. Modewise Validation and Cutoff Wavenumber Precision

The numerical performance of the regularized VNMM 2D ($\mathcal{P}^1$, $s_{\text{div}} = 6.0$, $\widehat{\text{Tol}} = 0.80$) for the first 10 physical TE_{nm} cavity modes is detailed in Table 8.

The regularized VNMM 2D achieves a global mean cutoff wavenumber error of 4.01% (maximum 7.33% for TE_{03}), preserving analytical degenerate pairings (TE_{10}/TE_{01}, TE_{20}/TE_{02}, TE_{21}/TE_{12}, TE_{30}/TE_{03}) with minor directional splitting ($\leq 0.5\%$) from the alternating director lattice and zero spurious modes in the operational band ($\lambda \leq 10$).

As reported in the final column of Table 8, the discrete divergence-to-curl energy quotient R_{div} remains bounded between 0.21% and 1.65% (mean $0.92\% \ll 1$), confirming that physical modes incur negligible penalty stiffening. In contrast, unregularized gradient modes exhibit $R_{\text{div}} \sim 10^3\%$–$10^6\%$; activating $s_{\text{div}} = 6.0$ selectively elevates these states ($\lambda_{\text{spur}} \geq 11.84$), shifting

Table 8: Comparison of the first 10 physical TE_{nm} resonant modes in the square PEC cavity against analytical references (361 active DoFs, 20×20 cells, 3×3 Gauss points).

Mode (TE_{nm})	$\lambda_{\text{analytical}}$	$k_{c,\text{analytical}}$	λ_{VNMM}	$k_{c,\text{VNMM}}$	**Error** λ (%)	**Error** k_c (%)	R_{div} (%)
TE_{10}	1.00	1.000	1.0081	1.004	0.81%	**0.41**%	0.22%
TE_{01}	1.00	1.000	1.0111	1.006	1.11%	**0.55**%	0.21%
TE_{11}	2.00	1.414	2.0809	1.443	4.05%	**2.00**%	0.71%
TE_{20}	4.00	2.000	4.2292	2.057	5.73%	**2.83**%	0.66%
TE_{02}	4.00	2.000	4.2505	2.062	6.26%	**3.08**%	0.70%
TE_{21}	5.00	2.236	5.5033	2.346	10.07%	**4.91**%	1.30%
TE_{12}	5.00	2.236	5.5255	2.351	10.51%	**5.12**%	1.50%
TE_{22}	8.00	2.828	9.1432	3.024	14.29%	**6.91**%	1.65%
TE_{30}	9.00	3.000	10.2889	3.208	14.32%	**6.92**%	1.33%
TE_{03}	9.00	3.000	10.3671	3.220	15.19%	**7.33**%	1.46%
Global Mean	—	—	—	—	**8.23**%	**4.01**%	**0.92**%

them past the tenth physical mode while leaving physical resonances unperturbed.

5.5.4. Parametric Sensitivity to the Div-Curl Penalty Parameter (s_{div})

Table 9 details the spectral response of the base cavity discretization ($N_x \times N_y = 21 \times 21$, 361 active DoFs) across penalty values $s_{\text{div}} \in [0.0, 50.0]$, delineating three distinct regimes:

1. **Under-Penalization ($s_{\text{div}} < 6.0$):** Unregularized formulation ($s_{\text{div}} = 0.0$) suffers static gradient nullspace leakage, contaminating the spectrum with spurious modes ($\lambda \sim 10^{-3}$). Increasing s_{div} stiffens these branches, shifting λ_{spur} upward ($0.00 \to 7.93$).
2. **Calibrated Operating Window ($s_{\text{div}} \in [6.0, 10.0]$):** At $s_{\text{div}} = 6.0$, the lowest spurious eigenvalue shifts beyond the tenth physical cavity mode ($\lambda_{\text{spur}} = 11.84 > 10.37$). No spurious modes contaminate the operational window ($\lambda \leq 10$), maintaining well-conditioned matrices ($\text{cond}_2 \sim 3 \times 10^3$) and bounded divergence residuals ($R_{\text{div}} \leq 1.65\%$).
3. **Over-Penalization ($s_{\text{div}} > 10.0$):** Excessive penalization amplifies discrete divergence residuals ($\nabla \cdot \vec{E}^h \sim \mathcal{O}(h)$), inducing volumetric stiffening, modest accuracy loss ($4.01\% \to 18.50\%$), and elevated conditioning ($\text{cond}_2 \sim 2.3 \times 10^4$ at $s_{\text{div}} = 50.0$).

Theoretical Scaling and A Priori Calibration Criterion for s_{div}. The scaling of the lowest spurious eigenvalue can be directly derived from the continuous

Table 9: Parametric sensitivity of cavity spectral purity, cutoff wavenumber error, and stiffness matrix conditioning to the div-curl penalty parameter s_{div} (361 active DoFs, 20×20 cells, 3×3 Gauss points).

s_{div}	**Spurious Modes**	**Lowest Spurious**	**Cutoff Error k_c**		$\text{cond}_2(\mathbf{K}_{\text{red}})$	**Max R_{div}**
	in Band (≤ 10)	**Eigenvalue** (λ_{spur})	**Mean (%)**	**Max (%)**		**Physical (%)**
0.0	10	0.00	96.57%	97.20%	3.10×10^5	—
1.0	5	1.99	11.13%	29.38%	7.31×10^2	9.09%
2.0	1	3.98	3.94%	19.33%	1.21×10^3	2.78%
4.0	1	7.93	2.03%	5.55%	2.15×10^3	1.98%
6.0	**0**	**11.84**	**4.01**%	**7.33**%	3.09×10^3	**1.65**%
8.0	0	15.78	4.92%	8.72%	4.03×10^3	1.45%
10.0	0	19.67	5.78%	10.03%	4.96×10^3	1.36%
20.0	0	38.97	9.54%	15.92%	9.54×10^3	1.22%
50.0	0	> 50	18.50%	29.67%	2.26×10^4	1.03%

regularized variational problem (53). For pure gradient fields $\vec{E} = \nabla\phi$ with homogeneous tangential Dirichlet conditions ($\phi = 0$ on $\partial\Omega$), the curl vanishes identically ($\nabla \times \vec{E} \equiv \mathbf{0}$), reducing the continuous regularized operator to:

$$-s_{\text{div}}\nabla(\nabla \cdot \vec{E}) = -s_{\text{div}}\nabla(\nabla^2\phi) = s_{\text{div}}\mu\nabla\phi = s_{\text{div}}\mu\vec{E}, \tag{74}$$

where μ denotes the eigenvalues of the scalar Dirichlet Laplacian ($-\nabla^2\phi = \mu\phi$ in Ω, $\phi = 0$ on $\partial\Omega$). For the square cavity $\Omega = [0, \pi]^2$, the Dirichlet Laplacian eigenvalues are $\mu_{m,n} = m^2 + n^2$ ($m, n \geq 1$), with the fundamental eigenvalue $\mu_{1,1} = 2.0$. Consequently, the lowest non-trivial gradient mode scales analytically as:

$$\lambda_{\text{spur}} \approx \mu_1 \cdot s_{\text{div}} = 2.0 \cdot s_{\text{div}}. \tag{75}$$

Under the present converged baseline quadrature (20×20 cells, 3×3 Gauss points), numerical evaluation confirms this scaling law down to 1.3% error ($\lambda_{\text{spur}}/s_{\text{div}} \approx 1.97$–$1.99$).

Equation (75) provides a predictive *a priori* calibration rule: ensuring complete spectral clearance up to a target physical eigenvalue $\lambda_{\max}$ (here $\lambda_{\max} \approx 10.4$ for the first 10 cavity modes) requires:

$$s_{\text{div}} > \frac{\lambda_{\max}}{\mu_1} = \frac{10.4}{2.0} = 5.2, \tag{76}$$

justifying the calibrated threshold $s_{\text{div}} \geq 6.0$ from continuous operator theory.

5.5.5. Sensitivity to Background Cell Density and Gauss Quadrature Order

To assess quadrature dependency, Table 10 reports the sensitivity across background grid densities and Gauss orders for the base cavity (361 active DoFs) compared to an ultra-refined reference (40×40 cells, 4×4 Gauss points, $N_g = 25{,}600$).

Table 10: Sensitivity of cavity spectral accuracy, discrete matrix operators, and assembly time to background cell density and Gauss quadrature order (361 active DoFs).

Background Grid	**Gauss Rule**	**Gauss Pts**	**Wavenumber Error**		**Relative Matrix Difference**		**Assembly**
($N_{cx} \times N_{cy}$)	($p \times p$)	(N_g)	**Mean k_c (%)**	**Max k_c (%)**	$\frac{\|\mathbf{K}-\mathbf{K}_{\text{ref}}\|_F}{\|\mathbf{K}_{\text{ref}}\|_F}$	$\frac{\|\mathbf{M}-\mathbf{M}_{\text{ref}}\|_F}{\|\mathbf{M}_{\text{ref}}\|_F}$	**Time (s)**
10×10 (100 cells)	2×2 (4 pts)	400	**1.09**%	**2.16**%	1.36×10^{0}	9.33×10^{-1}	**0.02** s
10×10 (100 cells)	3×3 (9 pts)	900	1.65%	4.78%	5.22×10^{-1}	4.18×10^{-1}	0.06 s
10×10 (100 cells)	4×4 (16 pts)	1600	2.29%	5.93%	4.17×10^{-1}	4.65×10^{-1}	0.09 s
20×20 (400 cells)	2×2 (4 pts)	1600	2.35%	5.97%	3.16×10^{-1}	3.65×10^{-1}	0.08 s
20×20 (400 cells)	4×4 (16 pts)	6400	2.82%	6.31%	8.42×10^{-2}	5.31×10^{-2}	0.36 s
30×30 (900 cells)	4×4 (16 pts)	14400	2.66%	5.80%	8.87×10^{-2}	1.03×10^{-1}	0.83 s
40×40 (1600 cells)	2×2 (4 pts)	6400	2.78%	6.07%	7.20×10^{-2}	5.95×10^{-2}	0.36 s
40×40 (1600 cells)	4×4 (16 pts)	25600	2.94%	6.05%	Reference	Reference	1.55 s

The results in Table 10 clarify the interaction between background quadrature and spatial discretization error. On coarse under-integrated grids ($10 \times 10, 2 \times 2$, $N_g/N_{\text{DoF}} \approx 1.11$), numerical softening across intra-cell stencil transitions shifts eigenvalues downward, partially cancelling positive spatial discretization stiffening ($\lambda > \lambda_{\text{exact}}$) and creating an apparent error of 1.09% that conceals a 140-dimensional numerical kernel in $\mathbf{K}_{\text{curl}}$. Refining quadrature eliminates this artificial softening and kernel defect ($\dim\ker(\mathbf{K}_{\text{curl}}) = 0$), revealing the true, uncorrupted spatial discretization error floor of 2.94%–4.01% for cutoff wavenumbers. Consequently, the resolution-matched convention ($N_{cx} = N_{cy} = N_v - 1, 3 \times 3$ Gauss points, $N_g/N_{\text{DoF}} \approx 10$, with cell spacing $h_{\text{cell}} = h_{\text{nom}}$) is adopted throughout all multi-level refinement and irregular cloud campaigns (Sections 5.6 and 5.7), ensuring that observed rates reflect the vector basis approximation rather than quadrature artifacts.

5.6. Comparative Convergence: Pure VNMM 2D vs. Edge FEM

To validate the asymptotic convergence of the proposed Vector Nodal Meshless Method against the canonical standard in computational electromagnetics, this section evaluates the resonant frequencies of the Transverse Electric (TE_z) cavity across six successive discretization levels over $\Omega = [0, \pi]^2$:

1. **Pure Edge FEM (Whitney 1-Forms):** Canonical conforming finite element solver on structured triangular meshes [3], associating one circuital degree of freedom with each interior edge ($N_{\text{DoF}} = N_{\text{edges,int}}$).
2. **Pure VNMM 2D ($\mathcal{P}^1$ Complete Basis):** Meshless solver over Ω configured under the baseline operational conventions established in Table 3: nominal spacing $h_{\text{nom}} = \pi/(N_v - 1)$, staggered boundary nodes, alternating directors, $s_{\text{div}} = 6.0$, $\widehat{\text{Tol}} = 0.80$, and baseline 3×3 Gauss quadrature ($N_g/N_{\text{DoF}} \approx 10$, $K_{\text{avg}} \approx 6.5$).

5.6.1. Resolution Metric and Degree-of-Freedom Equivalence

In a two-dimensional structured triangular mesh, Euler's planar graph formula yields [42]:

$$E \approx 3V, \tag{77}$$

producing $N_{\text{DoF}} \approx 3(N-1)^2$ for an $N \times N$ vertex Edge FEM mesh, whereas VNMM assigns one degree of freedom per internal node ($N_{\text{DoF}} = (N_v - 2)^2 \approx V$). To ensure a fair comparison with matching DoF density ($\rho_{\text{node}} \approx \rho_{\text{edge}}$), we define the edge-equivalent nodal discretization parameter:

$$N_v = \text{round}\big(\sqrt{3}\,(N-1)\big) + 1, \tag{78}$$

and track convergence as a function of the characteristic degree-of-freedom spacing:

$$h_{\text{DoF}} = \sqrt{\frac{|\Omega|}{N_{\text{DoF}}}}. \tag{79}$$

5.6.2. Convergence and Accuracy Comparison

Table 11 and Fig. 8 report the convergence history across 6 successive mesh levels ($N \in \{9, 13, 17, 21, 25, 29\}$), together with the Rayleigh error decomposition into pure curl and penalty contributions ($\lambda = \lambda_{\text{curl}} + s_{\text{div}}\lambda_{\text{div}}$) and the irregular cloud performance.

Key numerical observations include:

- **Strict Monotonicity and Locality:** The scale-invariant criterion $|\det(\widehat{\mathbf{A}})| \geq 0.80$ preserves compact nearest-neighbor stencils ($K_{\text{avg}} \approx 6.5$) across all grid levels without parameter tuning, preventing support dilation as $h \to 0$. In tandem with converged baseline quadrature (3×3 Gauss points), both Edge FEM and VNMM converge strictly monotonically toward the continuum limit.

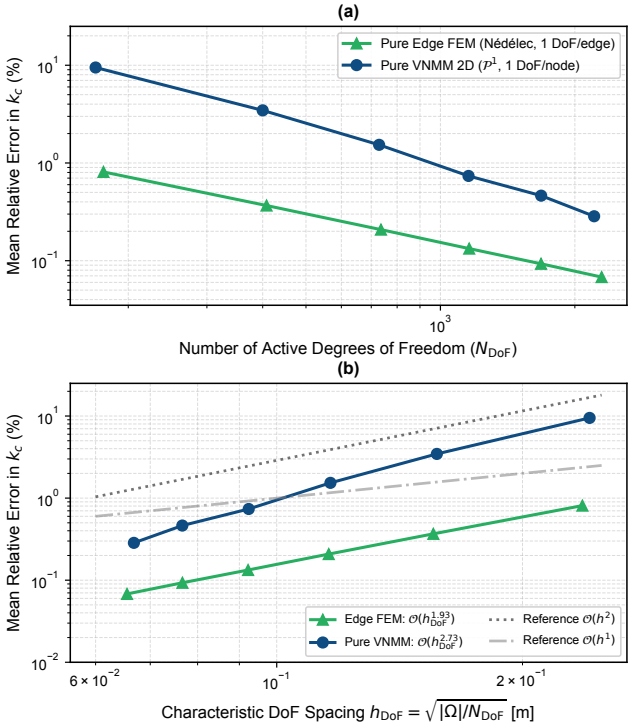


Figure 8: Spectral convergence comparison between Edge FEM and VNMM $\mathcal{P}^1$ for cavity TE_z modes. (a) Mean relative error in cutoff wavenumber k_c versus degrees of freedom N_{DoF}; (b) Convergence versus characteristic spacing h_{DoF}, showing fitted rates of $\mathcal{O}(h_{\mathrm{DoF}}^{1.93})$ (Edge FEM) and $\mathcal{O}(h_{\mathrm{DoF}}^{2.73})$ (VNMM).

Table 11: Comparative spectral convergence of resonant wavenumber k_c and error decomposition for cavity TE_z modes across six refinement levels ($s_{\mathrm{div}} = 6.0$, 3×3 Gauss quadrature): Edge FEM, VNMM $\mathcal{P}^1$ on regular grids (with Rayleigh curl and penalty biases), and VNMM $\mathcal{P}^1$ on irregular clouds (25% jitter, random directors).

Mesh	N_{DoF}	h_{DoF} [m]	**Edge FEM**	**VNMM $\mathcal{P}^1$ (Regular Grid)**				**VNMM $\mathcal{P}^1$ (Irregular)**
N			**Error (%)**	**Error k_c (%)**	**Error λ (%)**	**Curl Bias (%)**	**Penalty Bias (%)**	**Error k_c (%)**
9	169	0.2417	0.810	9.484	20.16	−4.78	+24.94	24.669
13	400	0.1571	0.367	3.454	7.08	+1.29	+5.79	11.489
17	729	0.1164	0.208	1.534	3.10	+0.66	+2.44	5.940
21	1156	0.0924	0.133	0.737	1.48	+0.03	+1.41	3.102
25	1681	0.0766	0.093	0.463	0.93	−0.89	+1.42	1.960
29	2209	0.0668	0.068	0.285	0.57	−1.14	+0.64	1.635

- **Error Hierarchy and Rates:** Edge FEM delivers lower absolute errors (0.81%–0.07%, rate $\mathcal{O}(h_{\mathrm{DoF}}^{1.93})$) due to exact circulation cancellation. VNMM displays an observed fitted slope of $\mathcal{O}(h_{\mathrm{DoF}}^{2.73})$ across all levels (9.48%–0.29%), steepening to $\mathcal{O}(h_{\mathrm{DoF}}^{2.97})$ on fine grids while operating entirely without topological mesh connectivity.

Pre-Asymptotic Penalty Relief and Rayleigh Decomposition. The observed fitted slope of $\mathcal{O}(h_{\mathrm{DoF}}^{2.73})$ reflects pre-asymptotic behavior governed by the interaction between curl and divergence penalty operators. As reported in Table 11, decomposing the discrete Rayleigh quotient into $\lambda = \lambda_{\mathrm{curl}} + s_{\mathrm{div}}\lambda_{\mathrm{div}}$ reveals that on coarse grids ($N = 9$), eigenvalue error (20.16%) is dominated by penalty stiffening (+24.94%), with pure curl contributing −4.78%. Under progressive refinement, discrete divergence rapidly vanishes ($\nabla \cdot \vec{E}^h \to 0$), causing the penalty bias to dissipate from +24.94% down to +0.64% at $N = 29$ ($\approx \mathcal{O}(h_{\mathrm{DoF}}^{2.57})$), while curl discretization error converges monotonically (−1.14%). This confirms that the steep slope represents pre-asymptotic relief of coarse-grid penalty stiffening coupled with dynamic stencil consolidation, narrowing the accuracy gap with Edge FEM from 11.7× to 4.2×.

5.7. Spectral Validation on Irregular Point Clouds with Arbitrary Vector Directors

To evaluate performance under simultaneous spatial and directional disorder, we discretize the cavity across the six refinement levels ($N_{\mathrm{DoF}} \in [169, 2209]$) using the baseline stochastic model (Table 3), maintaining the baseline quadrature and regularization settings (3×3 Gauss points, $s_{\mathrm{div}} = 6.0$, $\widehat{\mathrm{Tol}} = 0.80$).

Table 11 (final column) and Fig. 9 present the convergence history, while Table 12 details the first 10 physical modes on the fine cloud ($N_{\mathrm{DoF}} = 2{,}209$).

Critical findings from the irregular cloud campaign include:

1. **Monotonic Convergence under Disorder:** The regularized VNMM converges strictly monotonically with a fitted slope of $\mathcal{O}(h_{\mathrm{DoF}}^{2.20})$, confirming that spatial jitter and random directors do not cause spectral stagnation or locking, consistent with the completeness of $\mathcal{P}^1$.
2. **Error Hierarchy and Spectral Purity:** Coarse-mesh errors are elevated compared to regular grids (24.67% vs. 9.48% at $N = 9$) due to loss of Cartesian residual cancellation, but decay smoothly to 1.64% at $N = 29$. All 10 lowest computed eigenvalues match analytical physical

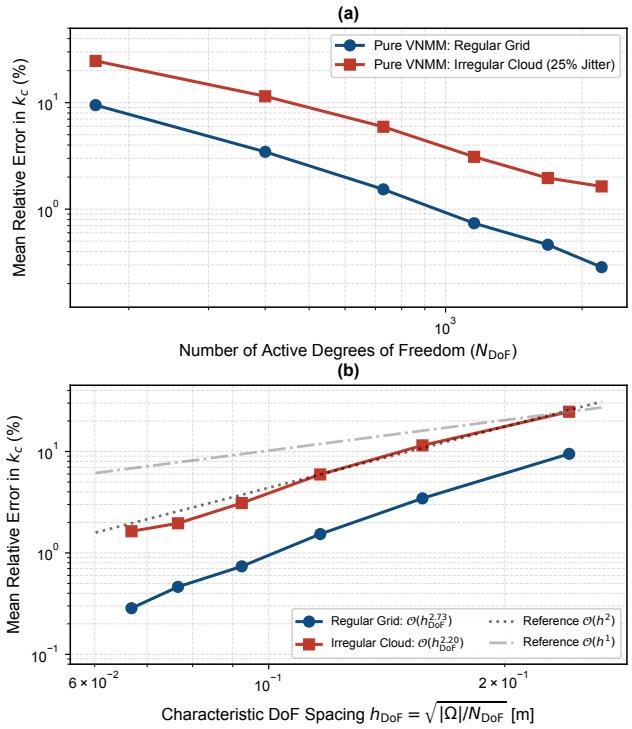


Figure 9: Spectral convergence of VNMM $\mathcal{P}^1$ on regular versus irregular point clouds. (a) Mean relative error in cutoff wavenumber k_c versus degrees of freedom N_{DoF}; (b) Convergence versus characteristic spacing h_{DoF}, showing fitted slopes of $\mathcal{O}(h_{\mathrm{DoF}}^{2.73})$ (regular grid) and $\mathcal{O}(h_{\mathrm{DoF}}^{2.20})$ (irregular cloud).

modes in exact 1-to-1 order, with zero spurious modes detected below $\lambda \leq 10$ (Table 12).

3. **Support Locality:** Enforcing $|\det(\widehat{\mathbf{A}})| \geq 0.80$ maintains compact candidate queries ($K_{\mathrm{avg}} \in [7.7, 8.2]$, $K_{\mathrm{max}} \leq 12$) across all levels, preventing stencil dilation despite severe spatial-directional disorder.

5.8. Limitations and Scope of Applicability

The numerical benchmarks confirm the stability and spectral accuracy of the stabilized VNMM 2D within the investigated canonical configurations. For broader CEM applications, four principal operational boundaries should be considered:

1. **Background Quadrature Scaffolding:** Evaluating the weak form currently relies on background Cartesian integration cells with 3×3

Table 12: First 10 physical TE_{nm} resonant cavity modes computed on the fine irregular point cloud against analytical references (2,209 active DoFs).

Mode (TE_{nm})	$\lambda_{\text{analytical}}$	λ_{VNMM}	**Error** λ **(%)**	$k_{c,\text{analytical}}$	$k_{c,\text{VNMM}}$	**Error** k_c **(%)**
TE_{10}	1.00	0.9511	4.89%	1.000	0.975	**2.48%**
TE_{01}	1.00	0.9544	4.56%	1.000	0.977	**2.31%**
TE_{11}	2.00	1.9424	2.88%	1.414	1.394	**1.45%**
TE_{20}	4.00	3.9493	1.27%	2.000	1.987	**0.64%**
TE_{02}	4.00	3.9930	0.17%	2.000	1.998	**0.09%**
TE_{21}	5.00	5.0366	0.73%	2.236	2.244	**0.37%**
TE_{12}	5.00	5.1316	2.63%	2.236	2.265	**1.31%**
TE_{22}	8.00	8.4144	5.18%	2.828	2.901	**2.56%**
TE_{30}	9.00	8.5648	4.84%	3.000	2.927	**2.45%**
TE_{03}	9.00	9.4952	5.50%	3.000	3.081	**2.71%**
Global Mean	—	—	**3.27%**	—	—	**1.64%**

Gauss quadrature to accommodate intra-cell stencil transitions (Section 5.5.5). While the Galerkin patch test demonstrated linear completeness for affine vector fields under this integration framework (Section 5.4), incorporating truly cell-free schemes (such as SCNI [43] or MLPG [15]) represents an essential step to eliminate background grids in complex domains.

2. **Boundary Stencil Starvation and Curvature:** On acute convex corners or curved boundaries, geometric truncation restricts the accessible angular half-space, which may require querying wider candidate neighborhoods ($K > 12$) to satisfy $|\det(\widehat{\mathbf{A}})| \geq \widetilde{\text{Tol}}$. Continuous boundary curvature also introduces tangential trace mismatch along curved facets unless boundary directors are adaptively aligned.
3. **High-Frequency Dispersion:** In electrically large domains ($kh \gg 1$), numerical dispersion requires dense spatial sampling ($h \leq \lambda_{\text{EM}}/10$), and the div-curl penalty parameter s_{div} may benefit from frequency-dependent scaling to maintain uniform spectral separation.
4. **Material Discontinuities and 3D Extension:** The present formulation focuses on 2D piecewise homogeneous domains. Discontinuous dielectric interfaces ($\epsilon_{r1} \neq \epsilon_{r2}$) require subdomain-partitioned stencils to prevent unphysical gradient smoothing, while 3D formulations require 12-node stencils ($\mathcal{P}_1^3$), increasing local combinatorial search costs.

6. Conclusion and Future Works

This work established a stabilized Vector Nodal Meshless Method (VNMM 2D) for electromagnetic vector field interpolation and eigenvalue analysis on unstructured point clouds without mesh connectivity. By resolving both local polynomial incompleteness and global variational instabilities, the proposed framework establishes the following core findings:

6.1. Summary of Findings and Methodological Conclusions

1. **Resolution of Local Incompleteness via $\mathcal{P}^1$:** Completing the local approximation space ($\mathcal{L}^1 \to \mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$) resolves all components of the first-order Jacobian $\nabla \vec{E}^h$, overcoming the $\mathcal{O}(1)$ curl noise floor in pointwise collocation. Furthermore, $\mathcal{P}^1$ unconditionally passes the Galerkin patch test to machine precision ($\leq 7.38 \times 10^{-14}$) for all seven affine modes—confirming linear completeness of the discrete Galerkin operator—and achieves systematic asymptotic convergence ($\mathcal{O}(h^{2.08})$ in L^2 and $\mathcal{O}(h^{0.75})$ in $H(\mathbf{curl})$) in the Galerkin Method of Manufactured Solutions (MMS), whereas the incomplete $\mathcal{L}^1$ basis exhibits catastrophic variational breakdown (15%–47% error) and complete convergence stagnation.
2. **Scale-Invariant Support Selection ($\widehat{\mathbf{A}} + k$-d tree):** Formulating support qualification via the dimensionless placement matrix $\widehat{\mathbf{A}}$ removes the quartic scaling $\mathcal{O}(h^4)$, decoupling stencil geometry from absolute grid refinement. Enforcing $|\det(\widehat{\mathbf{A}})| \geq \widehat{\text{Tol}}$ together with early-stopping k-d tree search preserves compact stencils ($K_{\text{avg}} \approx 6.5$–$11$) down to fine resolutions and bounds local conditioning ($\kappa_2(\widehat{\mathbf{A}}) \leq 5.9$) under 25% spatial disorder.
3. **Delineation of Topological Conformity vs. Mesh-Free Non-Conformity:** The formal proof of the Local Dual Equivalence Theorem (Appendix A) establishes that VNMM reproduces lowest-order Whitney edge elements on conforming simplices down to machine precision ($\approx 10^{-16}$). On arbitrary clouds, moving support stencils destroy global trace continuity ($V_h \not\subset H(\mathbf{curl}; \Omega)$) and discrete de Rham sequence exactness, causing gradient nullspace leakage in unregularized curl-curl formulations.
4. **Variational Div-Curl Regularization ($\mathbf{K}_{\text{curl}} + s_{\text{div}}\mathbf{K}_{\text{div}}$):** Volumetric div-curl regularization provides effective spurious mode shifting without expanding system dimensions. In the calibrated window

($s_{\text{div}} \in [6.0, 10.0]$), spurious gradient modes are shifted past the operational spectrum ($\lambda_{\text{spur}} = 11.84 > 10.37$) while physical resonances remain unperturbed ($R_{\text{div}} \leq 1.65\%$). Multi-level benchmarks show empirical refinement slopes of $\mathcal{O}(h_{\text{DoF}}^{2.73})$ on regular lattices and $\mathcal{O}(h_{\text{DoF}}^{2.20})$ under stochastic perturbations.

6.2. Promising Directions for Future Research

Building upon this 2D stabilized vector formulation, key avenues for future investigation include:

1. **3D Vector Formulations:** Extending the complete basis to 3D ($\mathcal{P}_1^3$, $M_{\text{supp}} = 12$) for 3D resonant cavities, waveguiding structures, and radiation problems [25].
2. **Cell-Free Quadrature and Complex Boundaries:** Replacing background integration cells with cell-free schemes (SCNI [43] or MLPG [15]) to eliminate quadrature scaffolding and accommodate moving or curved boundaries.
3. **Multi-Seed Statistical Characterization:** Conducting Monte Carlo campaigns across multiple random cloud realizations to evaluate statistical error distributions and qualification success rates.
4. **Material Interfaces:** Developing multi-tree spatial data structures [29] to restrict stencils to homogeneous subdomains, preserving constitutive jumps ($\epsilon_{r1} \neq \epsilon_{r2}$) and enforcing normal flux continuity across discontinuities.
5. **Hybrid EFEM–VNMM Coupling and Electrical Machines:** Coupling canonical Edge FEM in structured regions with VNMM in geometrically complex zones or narrow moving air gaps (e.g., in rotating electrical machinery [44]), leveraging the constructive Whitney equivalence for seamless interface continuity without remeshing or sliding-mesh interpolation artifacts.

Appendix A. Formal Proof of the Local Constructive Equivalence Theorem

This appendix provides the complete constructive proof of Theorem 1 (Local Constructive Equivalence under Conforming Midpoint/Tangent Placement between the VNMM $\mathcal{L}^1$ basis and lowest-order Whitney 1-forms on triangular simplices).

Proof. The proof proceeds in five constructive steps:

Step 1: Identity of Local Vector Polynomial Spaces. The lowest-order Whitney space $\mathcal{W}(T)$ on T is spanned by fields of the form $[a_1+\alpha y, a_2-\alpha x]^T$. By inspection, $\mathcal{W}(T) \equiv \mathcal{L}^1 = \text{span}\{[1,0]^T, [0,1]^T, [y,-x]^T\}$, so both formulations span the same 3-dimensional vector polynomial space.

Step 2: Constancy of Tangential Field Projection Along Simplex Edges. Let $\vec{u}(\mathbf{x}) = [a_1 + \alpha y, a_2 - \alpha x]^T \in \mathcal{L}^1$. The directional projection of $\vec{u}(\mathbf{x})$ along unit director $\vec{t}_k = [\Delta x_k/L_k, \Delta y_k/L_k]^T$ is:

$$\vec{u}(\mathbf{x}) \cdot \vec{t}_k = \frac{1}{L_k} \left[a_1 \Delta x_k + a_2 \Delta y_k + \alpha(y\Delta x_k - x\Delta y_k) \right]. \tag{A.1}$$

Since any point $\mathbf{x} = (x, y)$ on edge e_k satisfies $(x-x_k)\Delta y_k - (y-y_k)\Delta x_k = 0$, the rotational term $y\Delta x_k - x\Delta y_k = y_k\Delta x_k - x_k\Delta y_k$ is strictly constant along the entire edge length. Therefore, the tangential projection is invariant along e_k:

$$\vec{u}(\mathbf{x}) \cdot \vec{t}_k = \vec{u}(M_k) \cdot \vec{t}_k, \quad \forall \mathbf{x} \in e_k. \tag{A.2}$$

Step 3: Reduction of Edge Circulation to Pointwise Midpoint Value. Using Eq. (A.2), the line integral defining the FEM circulation degree of freedom along edge e_j simplifies to:

$$\oint_{e_j} \vec{u} \cdot d\boldsymbol{\ell} = \int_{-L_j/2}^{L_j/2} \left(\vec{u}(\mathbf{x}(s)) \cdot \vec{t}_j \right) ds = \left(\vec{u}(M_j) \cdot \vec{t}_j \right) \int_{-L_j/2}^{L_j/2} ds = L_j \left(\vec{u}(M_j) \cdot \vec{t}_j \right). \tag{A.3}$$

Substituting the k-th Whitney basis function $\vec{w}_k \in \mathcal{W}(T)$, which satisfies $\oint_{e_j} \vec{w}_k \cdot d\boldsymbol{\ell} = \delta_{jk}$ by construction, yields:

$$L_j \left(\vec{w}_k(M_j) \cdot \vec{t}_j \right) = \delta_{jk} \quad \Longrightarrow \quad (L_k \vec{w}_k(M_j)) \cdot \vec{t}_j = \delta_{jk}. \tag{A.4}$$

Step 4: Uniqueness and Exact Identification with VNMM Shape Functions. By definition of the VNMM 2D $\mathcal{L}^1$ interpolation problem on the support triad $\{M_j, \vec{t}_j\}_{j=1}^3$, the shape functions satisfy $\vec{N}_k(M_j) \cdot \vec{t}_j = \delta_{jk}$. Consider the difference field:

$$\vec{d}_k(\mathbf{x}) = \vec{N}_k(\mathbf{x}) - L_k \vec{w}_k(\mathbf{x}) \in \mathcal{L}^1. \tag{A.5}$$

Evaluating its projection at each midpoint M_j:

$$\vec{d}_k(M_j) \cdot \vec{t}_j = \vec{N}_k(M_j) \cdot \vec{t}_j - L_k \left(\vec{w}_k(M_j) \cdot \vec{t}_j \right) = \delta_{jk} - \delta_{jk} = 0, \quad \forall j \in \{1,2,3\}. \tag{A.6}$$

By Steps 2–3, $\vec{d}_k$ has zero circulation along every edge, and edge circulations are unisolvent on $\mathcal{W}(T) = \mathcal{L}^1$ for non-degenerate T; hence $\vec{d}_k \equiv \mathbf{0}$ (equivalently, the local moment placement matrix $\mathbf{A} \in \mathbb{R}^{3\times 3}$ is non-singular with $\det \mathbf{A} \neq 0$), which proves the exact identity $\vec{N}_k(\mathbf{x}) \equiv L_k \vec{w}_k(\mathbf{x})$ everywhere in $\mathbb{R}^2$.

Step 5: Verification of Differential Invariants. 1. *Divergence:* For any field $\vec{u} = [a_1 + \alpha y, a_2 - \alpha x]^T \in \mathcal{L}^1$, we have $\nabla \cdot \vec{u} = \partial(a_1 + \alpha y)/\partial x + \partial(a_2 - \alpha x)/\partial y \equiv 0$. Thus, $\nabla \cdot \vec{w}_k \equiv 0$ and $\nabla \cdot \vec{N}_k \equiv 0$ everywhere. 2. *Curl:* Applying Stokes' theorem over triangle T for Whitney 1-form $\vec{w}_k$:

$$\int_T (\nabla \times \vec{w}_k)_z d\Omega = \oint_{\partial T} \vec{w}_k \cdot d\boldsymbol{\ell} = \sum_{j=1}^{3} \delta_{jk} = 1 \implies (\nabla \times \vec{w}_k)_z = \frac{1}{|T|}. \quad \text{(A.7)}$$

By the identity proved in step 4, $(\nabla \times \vec{N}_k)_z = L_k(\nabla \times \vec{w}_k)_z = L_k/|T|$. □

Appendix B. Derivation of the Derivative Aliasing Scaling in the Incomplete Basis

This appendix details the order-of-magnitude scaling derivation supporting Heuristic 1.

Derivation of Scaling Estimate. Let any discrete vector field $\vec{E}^h \in \mathcal{L}^1$ be expanded in terms of the solenoidal basis:

$$\vec{E}^h(x, y) = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c \begin{bmatrix} y \\ -x \end{bmatrix}, \quad a, b, c \in \mathbb{R}. \quad \text{(B.1)}$$

Direct differentiation with respect to spatial coordinates yields the discrete Jacobian tensor:

$$\mathbf{J}^h = \begin{bmatrix} \frac{\partial E_x^h}{\partial x} & \frac{\partial E_x^h}{\partial y} \\ \frac{\partial E_y^h}{\partial x} & \frac{\partial E_y^h}{\partial y} \end{bmatrix} = \begin{bmatrix} 0 & c \\ -c & 0 \end{bmatrix}. \quad \text{(B.2)}$$

Inspection shows that $\partial E_x^h/\partial x \equiv 0$, $\partial E_y^h/\partial y \equiv 0$, and $\partial E_x^h/\partial y + \partial E_y^h/\partial x = c - c \equiv 0$, identically satisfying the linear constraints in Eq. (20).

Now consider a general smooth physical field $\vec{E} \in [C^2(\Omega)]^2$. Its Taylor expansion centered at evaluation point P is $\vec{E}(\mathbf{x}) = \vec{E}(P) + \mathbf{J}(P)\Delta\mathbf{x} + \mathcal{O}(h^2)$, where $\mathbf{J}(P)$ has non-zero diagonal entries ($\partial E_x/\partial x \neq 0$ or $\partial E_y/\partial y \neq 0$).

Collocating $\vec{E}^h$ at support nodes $\mathbf{x}_k$ $(k = 1, 2, 3)$ along unit directors $\vec{t}_k$ defines the linear placement system $\mathbf{A}[a, b, c]^T = \mathbf{e}$, where $e_k = \vec{E}(\mathbf{x}_k) \cdot \vec{t}_k$.

Subtracting the projection of the unrepresented linear components $(\mathbf{J}(P) - \mathbf{J}^h)\Delta\mathbf{x}_k \cdot \vec{t}_k$ yields a nodal projection residual $\mathbf{r} \in \mathbb{R}^3$ with asymptotic magnitude $\|\mathbf{r}\| \sim \mathcal{O}(h)$. In the 3×3 moment placement matrix $\mathbf{A}$ (Eq. (11)), translation columns 1–2 scale as $\mathcal{O}(1)$, while the rotational moment column 3 scales as $(\Delta y_k t_{kx} - \Delta x_k t_{ky}) \sim \mathcal{O}(h)$. Consequently, by Cramer's rule or Neumann inversion, the third row of $\mathbf{A}^{-1}$ (which maps nodal data to the rotational DoF $c = -\frac{1}{2}(\nabla \times \vec{E}^h)_z$) scales strictly as $\| \operatorname{Row}_3(\mathbf{A}^{-1})\| \sim \mathcal{O}(h^{-1})$.

The numerical error in the reconstructed curl at evaluation point P is therefore governed by the propagation of the projection remainder:

$$\begin{aligned} |(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z| &= 2|c_{\text{exact}} - c| \\ &\sim \| \operatorname{Row}_3(\mathbf{A}^{-1})\|\|\mathbf{r}\| = \mathcal{O}(h^{-1}) \cdot \mathcal{O}(h) = \mathcal{O}(1), \end{aligned} \tag{B.3}$$

which is asymptotically independent of h and generates an $\mathcal{O}(1)$ error plateau under spatial refinement ($h \to 0$) in the collocated setting for generic node/director placements (see Remark 1). □

CRediT Authorship Contribution Statement

Renato C. Mesquita: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Funding acquisition. **Elson J. da Silva**: Conceptualization, Formal analysis, Writing – review & editing. **Bárbara M. F. Gonçalves**: Methodology, Software, Validation, Investigation, Formal analysis, Writing – review & editing. **Luilly A. G. Ortiz**: Methodology, Software, Validation, Data curation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The simulation scripts, benchmark datasets, and point-cloud generation algorithms supporting the findings of this study are available from the corre-

sponding author upon reasonable request.

Acknowledgements

The authors would like to thank the Brazilian research funding agencies CAPES, CNPq, and FAPEMIG for their financial support during the development of this research project.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used Google Antigravity / Gemini in order to assist with English language editing, LaTeX code formatting, script automation for numerical data consistency, and cross-referencing verification. After using this tool, the authors carefully reviewed, verified, and edited all resulting content and take full responsibility for the integrity and accuracy of the publication.

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